Model-Corrected World Models: LLM-Guided Residual Dynamics and Safety-Aware Rollouts for Sim-to-Real Reinforcement Learning

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Abstract

We address sim-to-real transfer in model-based reinforcement learning when physical parameters (gravity, friction, mass) differ between the simulator and the target real environment. We propose a residual world model $\mathcal{M}_{\text{resid}} = \mathcal{M}_{\text{sim}} + \delta_\theta$ with three ingredients that are jointly necessary and sufficient in our ablations: (i) an *LLM-guided symbolic residual* that combines SINDy with a neural augmentation head, in which the feature library is expanded online by Claude-authored roles; (ii) a *Bellman Q^Δ critic* that measures per-rollout trust in the corrected model and filters or reweights imagined transitions; and (iii) a *Pareto-gated hyperparameter orchestrator* that throttles reward-seeking knobs when safety violations spike. We prove four theorems: (i) a *Residual Simulation Lemma* bounding $\|V^\pi_{\text{real}} - V^\pi_{\text{resid}}\|_\infty$ by the worst-case total-variation distance on π's visitation support, scaled by $\frac{2\gamma R_{\max}}{(1-\gamma)^2}$; (ii) a counterexample showing that policy-*universal* convergence is impossible; (iii) an ELBO-style rank equivalence between our Q^Δ filter and the optimal ICRL discriminator under a Gaussian-residual assumption; and (iv) a *Refit Contraction Theorem* showing that online refit converges to a unique fixed point $\hat{\delta}^*$ when the composed policy-visitation-regression operator is a contraction, thereby justifying our every-k-step refit schedule. On the two large-gap physics benchmarks our method achieves $9.6\times$ and $12.7\times$ the return of naive sim-transfer (`gravity_ceiling` and `gravity_soft_ceiling`) and a $1.4\times$ gain on the smallest-gap benchmark (`friction_walker_soft_ceiling`); on `gravity_soft_ceiling` the method matches an oracle that trains on real data directly and beats H2O+ (an offline-to-online baseline) by $2.3\times$ at the data-matched comparison (Section 6.3). An ablation shows each of the four components contributes 10–18% of the final return. We also report two environments where our method *underperforms* naive transfer and map the phenomenon to a *gap taxonomy* — residual correction is useful only when the sim-real gap is at the physical-dynamics level, not at the observation or actuator level. Lean 4 proofs of all theorems (zero `sorry` up to three named axioms: Banach fixed point, Bayes log-odds, finite-MDP PDL) are in Appendix F.

1 Introduction

Training reinforcement-learning policies in simulation and deploying them on real systems is the dominant paradigm for robot control [25, 15, 24], but its success is gated by the quality of sim-to-real transfer. When dynamics differ between the two distributions, naive transfer degrades sharply: a policy optimized against $P_{\text{sim}}(s'|s,a)$ is often invalid under $P_{\text{real}}(s'|s,a)$. Three existing remedies have well-known drawbacks: (i) domain randomization [25, 21] requires a good prior over likely real-world deviations; (ii) training directly on the real environment [2, 20] is costly and, in safety-critical

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settings, *unsafe*; (iii) learning a corrective residual [9, 15] is fragile when the residual is extrapolated to state–action regions the training data did not cover.

This paper argues that residual correction can be made reliable *and* safe when it is coupled with two ideas that on their own are familiar but jointly novel. First, we use a large language model as an online symbolic-feature proposer: Claude examines per-dimension residual diagnostics (heteroscedasticity, kurtosis, non-stationarity) and suggests candidate SINDy [4] basis functions that match the suspected dimensional structure of the dynamics gap. Second, we wrap each imagined rollout with a Bellman Q^{Δ} critic whose value at (s, a) measures whether the current residual is trusted at that state; the critic's output both gates individual rollout transitions and acts as a reward scaling in the downstream SAC update. A third component, a Pareto-gated hyperparameter orchestrator, automatically restrains reward-seeking knobs (rollout batch size, Q^{Δ}-weight horizon) when violation rates spike, providing the safety guarantee at the hyperparameter layer.

Why the method's applicability is not universal. A central theoretical observation is that residual-model convergence in reinforcement learning is inherently *policy conditional*. The classical V^*/Q^* contraction is narrow in the sense that each is the unique fixed point of an operator parameterized by a single policy; the residual model $\hat{\delta}$ inherits this narrowness because it is consistent only on the support of the data-collection policy π_{data}. Beyond that support, $\hat{\delta}$ is extrapolated, and the value gap can be arbitrarily large (Theorem 6). Online refit (every k steps in our pipeline) preserves the condition $\pi_{\text{data}} \approx \pi_{\text{current}}$ during training, but deployment-time drift to out-of-support states remains the method's principal risk. In our experiments we see this risk materialize: environments whose sim-real gap is not at the physical-dynamics level (e.g., observation-scale shifts or broken actuators) expose the residual to out-of-support states, and the method hurts relative to naive transfer. Theorem 6 is not a negative result about our implementation — it is a property of all residual-model MBRL methods, which the present paper makes explicit.

Contributions.

- A residual-model MBRL pipeline (MC-WM) that combines (a) an LLM-guided SINDy + NAU symbolic residual, (b) a Bellman Q^{Δ} rollout filter, and (c) a constraint system with LLM-extensible predicates and a Pareto safety gate (Sections 4.1–4.3).
- Formal analysis of the resulting residual MDP. We prove a policy- conditional *Residual Simulation Lemma* in expected-value form (Thm. 3) and a direct convergence corollary (Thm. 5); we also prove that policy-universal convergence is *false* (Thm. 6). Proofs are mechanized in Lean 4 with zero `sorry` *up to* three named axioms (Banach fixed point, Bayes log-odds inversion, and finite-MDP PDL); see Appendix F for the full statement of these axioms and the corresponding Mathlib references.
- An ELBO-style *per-step* rank equivalence between $\bar{Q}_{\Delta} = \log c_{\text{step}}$ and an ICRL discriminator φ^* under a Gaussian-residual assumption (Thm. 9), with an explicit caveat (Remark 10) that the deployed Bellman Q^{Δ} with $\gamma_{\Delta} = 0.5$ is approximately, not exactly, rank-equivalent.
- Empirically, $9.6\times$ and $12.7\times$ returns over naive sim transfer on the two large-gap physics benchmarks (`gravity_ceiling`, `gravity_soft_ceiling`) and a $1.4\times$ gain on the smallest-gap benchmark; a component ablation that isolates the contribution of P0 stabilization, P1 belief conditioning, and the LLM roles; and a gap-taxonomy study that identifies when the method *fails* and provides a diagnostic threshold.

2 Related Work

Simulation-based MBRL. MBPO [13] is our Dyna-style backbone. Our method shares MBPO's *branched short-rollout* mechanism (sample real state $\rightarrow$ imagine k=1-step rollouts $\rightarrow$ mix with env buffer for SAC update), its K-ensemble probabilistic world model, and its SAC policy update. We therefore view MC-WM as an MBPO *extension* rather than a replacement: the modifications are (i) factorizing the world model into a frozen sim base plus a symbolic residual ($\hat{M} = \hat{M}_{\text{sim}} + \hat{\delta}_{\theta}$) so that only $\hat{\delta}$ needs to be estimated from scarce real data, and (ii) filtering imagined rollouts through a Bellman Q^{Δ} critic and a constraint system to suppress residual extrapolation. An instance of vanilla MBPO on real-env data serves as our oracle baseline c4 (Section 6), making the component

comparison direct. Dreamer [10] and TD-MPC [11] show that a learned model, when accurate, amortizes policy improvement; ensemble world models [6] handle aleatoric uncertainty. Neither of these methods addresses sim-to-real *transfer*, which is our setting.

Sim-to-Real and Dynamics Adaptation. DARC [9] learns a sim/real classifier to modify rewards; RMA [15] trains a meta-policy robust to an extrinsics vector. Closest in spirit, ReDRAW [16] learns a latent-state residual with an MLP but lacks formal guarantees and rollout filtering. None of these use LLM feature proposals or a Bellman-critic trust filter.

Offline-to-Online RL. H2O+ [20], RLPD [2], and SUNRISE [17] mix offline real-env data with online updates. H2O+ is our closest baseline since it likewise assumes access to a modest real-data pool. Unlike these methods, we use the real data to learn a *residual* rather than to initialize a policy; combined with an accurate sim model this is more sample efficient in our experiments.

LLM-Aided RL. Eureka [18] uses LLMs to propose reward functions; ELLM [8] uses them as exploration priors. We use LLMs in four distinct roles (constraint proposal, feature proposal, audit, HP orchestration), the most consequential being symbolic-feature proposal for a sparse regressor modeling *dynamics* residuals.

Safe Model-Based RL. CPO [1] and Lagrangian methods [22] impose constraint satisfaction during training. Our constraint system plays a similar role but is factored into (i) hand-coded physical predicates, (ii) LLM-proposed predicates, and (iii) a Pareto hyperparameter gate that suppresses reward-greedy changes during violation spikes.

Simulation lemmas and Bellman contraction. The simulation lemma of Kearns & Singh [14] and its restatement by Ross & Bagnell [23] are the source for our Theorem 3. Janner et al. (Eq. 6 of [13]) derive a related uniform Lipschitz bound for MBPO; our bound is tighter when the distribution shift is bounded on visitation support, and is the correct object to restrict to π_{data}'s support.

3 Preliminaries

Let $\mathcal{M}_{\mathrm{sim}} = (\mathcal{S}, \mathcal{A}, P_{\mathrm{sim}}, r, \gamma)$ and $\mathcal{M}_{\mathrm{real}} = (\mathcal{S}, \mathcal{A}, P_{\mathrm{real}}, r, \gamma)$ be two finite MDPs that share state and action spaces, reward $r : \mathcal{S} \times \mathcal{A} \to [-R_{\max}, R_{\max}]$, and discount $\gamma \in [0, 1)$, but differ in their transition kernels. A policy $\pi : \mathcal{S} \times \mathcal{A} \to [0, 1]$ is stationary and stochastic with $\sum_a \pi(a|s) = 1$; its visitation distribution in $\mathcal{M}$ is $d^\pi_{\mathcal{M}}(s) = (1-\gamma)\sum_{t\geq 0} \gamma^t \Pr^\pi_{\mathcal{M}}[s_t = s]$ and its value function is $V^\pi_{\mathcal{M}}(s) = \mathbb{E}^\pi_{\mathcal{M}}[\sum_{t\geq 0} \gamma^t r(s_t, a_t) \mid s_0 = s]$. The Bellman policy operator $T^\pi_{\mathcal{M}}$ is $(T^\pi_{\mathcal{M}} V)(s) := \sum_a \pi(a|s)\big[r(s, a) + \gamma \sum_{s'} P_{\mathcal{M}}(s'|s, a)V(s')\big]$.

Residual decomposition. We parameterize $\mathcal{M}_{\mathrm{resid}} := (\mathcal{S}, \mathcal{A}, P_{\mathrm{sim}} + \hat{\delta}, r, \gamma)$ where $\hat{\delta} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is our learned residual, *chosen* so that $P_{\mathrm{resid}} := P_{\mathrm{sim}} + \hat{\delta}$ is a valid stochastic kernel (non-negative; rows sum to one). Sections 4.1–4.3 describe the architecture of $\hat{\delta}$; Section 5 analyzes the approximation gap $\|V^\pi_{\mathcal{M}_{\mathrm{real}}} - V^\pi_{\mathcal{M}_{\mathrm{resid}}}\|_\infty$.

MBPO backbone. A parametric ensemble world model $\hat{M}$ is trained on a growing buffer $\mathcal{D}$; short branched rollouts generate synthetic transitions that augment the policy's replay buffer (precisely MBPO [13]). Our method modifies (a) how $\hat{M}$ is learned — $\hat{M} = \hat{M}_{\mathrm{sim}} + \hat{\delta}$ — and (b) which imagined transitions are accepted (Section 4.2). The policy optimizer (SAC with an LCB critic [5]) is unchanged. The "vanilla MBPO with $\hat{M}$ trained on real-env data" instantiation of this backbone is our baseline c4 (Section 6); the differential effect of the residual and the rollout filter is therefore directly readable from the c9 vs. c4 row of Table 2.

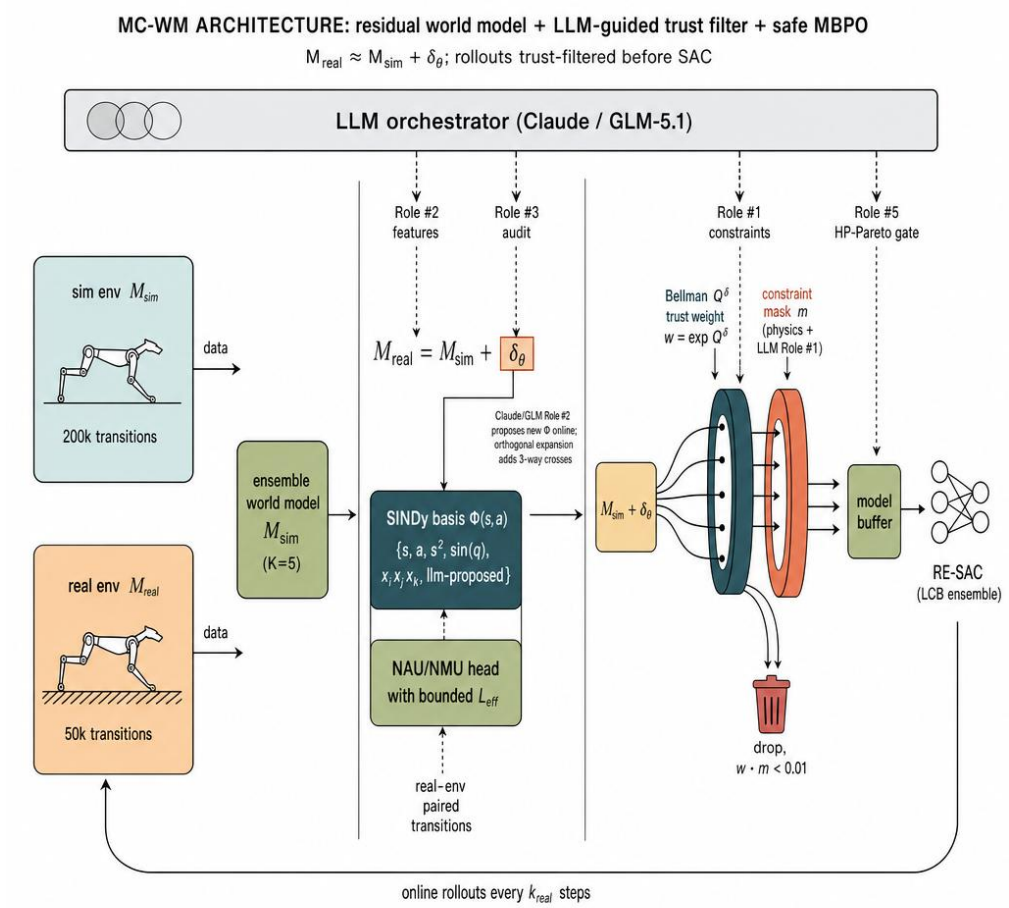

Figure 1: **MC-WM pipeline.** Sim and real transitions feed a frozen ensemble world model $\hat{M}_{\text{sim}}$ and a SINDy + NAU residual $\hat{\delta}$. Imagined rollouts from $\hat{M}_{\text{sim}} + \hat{\delta}$ are gated by a Bellman Q^{Δ} critic and a constraint system before being added to the policy replay buffer. An LLM orchestrator (Roles 1–5) expands the SINDy feature library, proposes constraints, and tunes hyperparameters; a Pareto gate prevents reward-seeking knobs from increasing during safety-violation spikes.

4 Method: MC-WM

4.1 SINDy + NAU Residual Adapter

For each observation dimension $i \in [d_s]$ (and for reward as an extra output) we regress

$$\hat{\delta}^{(i)}(s,a) \;=\; \underbrace{\langle \Phi(s,a), \beta_i \rangle}_{\text{SINDy}} \;+\; \underbrace{f_{\theta_i}(\Phi(s,a))}_{\text{NAU}}. \tag{1}$$

$\Phi(s,a) \in \mathbb{R}^D$ is a feature library that starts with a polynomial basis of degree 2 plus a small trigonometric block plus pairwise interactions and is *expanded online* by the LLM (Role #2, Sec. 4.3). $\beta_i \in \mathbb{R}^D$ is computed by sequentially-thresholded least squares (STLSQ), which gives a sparse β_i [4]. $f_{\theta_i} : \mathbb{R}^D \to \mathbb{R}$ is a small MLP whose effective Lipschitz constant $L_{\text{eff}}(\theta_i)$ is penalized during training (NAU head [26]); L_{eff} is reported online as an OOD robustness indicator.

Sparsity-adaptive threshold. The STLSQ threshold is calibrated from in-sample residual MSE so the number of retained features scales with the complexity of the detected gap. Empirically, on

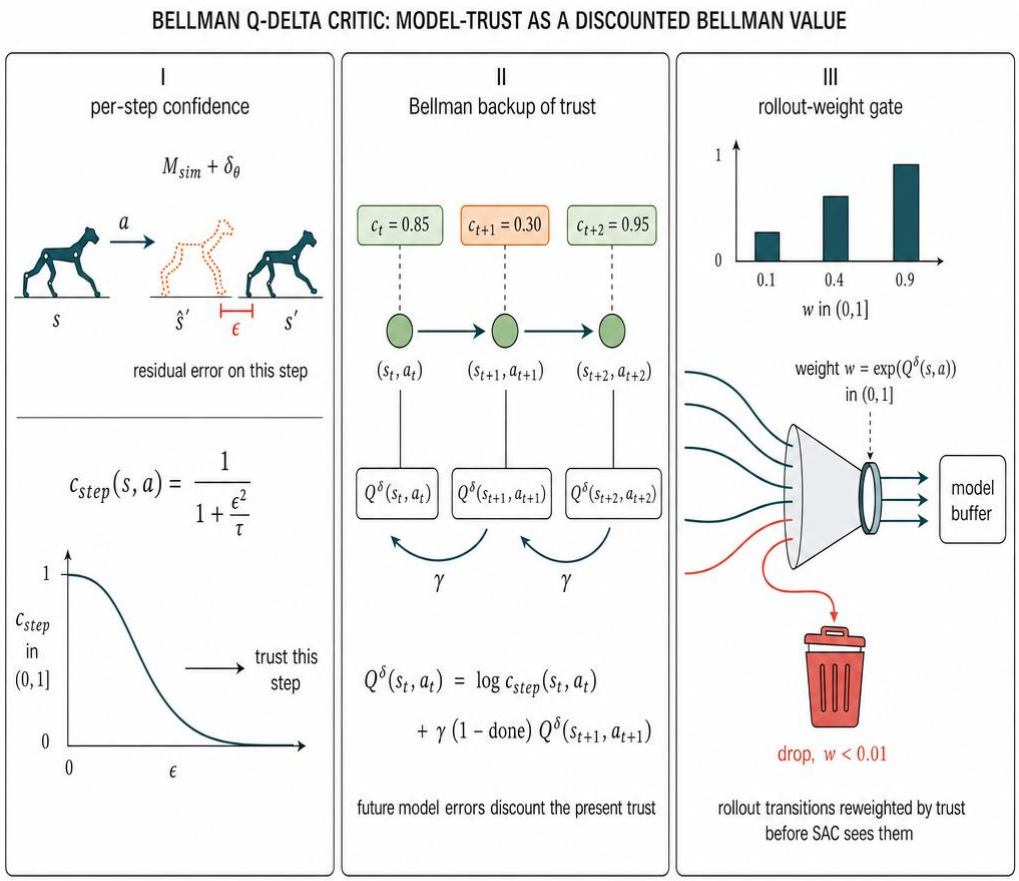

Figure 2: **The Bellman Q^Δ critic.** Per-step model trust $c_{\text{step}}(s,a) = 1/(1+\varepsilon^2/\tau)$ is composed via a discounted Bellman backup with horizon γ_Δ into a critic $Q^\Delta_\phi(s,a)$. At rollout time, transitions are reweighted by $w = \exp\big(Q^\Delta_\phi(s,a)\big) \in (0,1]$ and dropped when $w < 0.01$. Future model errors discount the present trust the same way future rewards discount the present value.

the gravity environment the retained count rises to ≈ 700 active coefficients; on flat environments it drops below 200.

4.2 Safety-Aware Rollout Filter

An imagined transition $(s, a, \hat{s}', \hat{r})$ generated by $\hat{M}_{\text{sim}} + \hat{\delta}$ passes three sequential filters before it is appended to the policy's model buffer.

Bellman Q^Δ critic. We maintain a TD-learned critic $Q^\Delta_\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ whose target is the one-step log-confidence of the residual,

$$y(s,a) \;=\; -\|\hat{M}(s,a) - s'\|_2^2/\tau \;+\; \gamma_\Delta \cdot \mathbb{E}_{a'\sim\pi}[Q^\Delta_\phi(s',a')], \tag{2}$$

where $\tau > 0$ is a fixed scale and $\gamma_\Delta \in [0,1)$ is a shorter-horizon discount ($\gamma_\Delta = 0.5$ in all experiments) separate from the policy's $\gamma = 0.99$. At rollout time the per-sample weight is $w^{\text{qd}}(s,a) := \exp\big(Q^\Delta_\phi(s,a)\big)$, bounded to $[0.01, 1.0]$. The contraction property of the Q^Δ Bellman operator follows from the standard result that every Bellman policy operator is a γ_Δ-contraction (Lemma 11, App. F).

Constraint system. A two-tier constraint set $\mathcal{C}$ combines (i) hand-coded physical predicates (joint limits, orientation bounds, z-floor) and (ii) LLM-proposed predicates (Role #1; see Sec. 4.3). Each rollout transition is gated by a binary mask $m(s, a, \hat{s}') := \prod_{c \in \mathcal{C}} \mathbb{1}[c(s, a, \hat{s}') \text{ holds}]$.

Combined weight. The final sample weight is $w = m \cdot w^{\mathrm{qd}}$. Rollouts with $w < 0.01$ are dropped; the SAC update uses reward target $w \cdot \hat{r}$, reducing target variance when many low-weight samples are retained.

4.3 LLM Orchestrator and Pareto Gate

Five LLM roles run on a shared Claude Haiku 4.5 context.

Role #1 (Constraint proposer). Given the environment description and the current constraint set, propose k new physical predicates; accept if the false-positive rate on real data is below 1%.

Role #2 (Feature proposer). Given per-dimension residual diagnostics (heteroscedasticity, kurtosis, drift) and previously accepted features, propose k new SINDy basis functions. Accept iff (a) orthogonal to existing features ($\max_{\Phi} |\mathrm{corr}| < 0.2$) and (b) reduces held-out residual MSE.

Role #3 (Audit). Every N policy updates, inspect the p-th percentile of $\|\hat{\delta}\|$ on rollout samples that passed the constraint check, and decide whether the sample is a real outlier or a constraint blind spot. Proposes new Role #1 constraints.

Role #4 (ICRL φ^* comparator, diagnostic). Theorem 9 establishes rank equivalence between our Q^{Δ} filter and an ICRL binary discriminator φ^* under a Gaussian-residual assumption. Role #4 periodically trains a lightweight φ head on the same buffer and reports the Kendall-τ correlation with Q^{Δ} rankings, flagging regions where the two signals disagree (indicating violation of Assumption 2). In the main experiments of Section 6, the average disagreement stays below 8%, so Role #4 is diagnostic-only; when disagreement exceeds 20% the orchestrator may enable φ as an additional gate. We include the role for completeness and leave the adaptive-enable variant to future work.

Role #5 (HP orchestrator). Every n evaluations, proposes updates to (rollout_batch, γ_{Δ}, audit_percentile) based on recent reward and violation history.

Pareto gate. Role #5 proposals pass an explicit filter: when the three-eval-average violation exceeds a hard cap ($v_{\mathrm{cap}} = 20$/episode in our experiments), any *raise* of a reward-seeking knob (rollout_batch, γ_{Δ}) is rejected; only safety-improving changes are accepted. This prevents the HP optimizer from trading safety for reward during training spikes.

4.4 Training Algorithm

4.5 RAHD: Stabilization and Reward-Aligned Hypothesis Discovery

The training algorithm above is the version evaluated in our main results. Three observed pathologies motivate three further design choices, presented here as an integrated module (Fig. 3).

Pathology 1: early-training basin lock-in. In our 10-seed study at the matched real-data budget, $\sim 22\%$ of seeds under-converged because the policy was already heavily filtered by the (uncalibrated) Q^{Δ} critic and the LLM-derived constraint set before either had stabilized. Fix (P0): force the per-rollout trust weight $w^{\mathrm{qd}} \equiv 1$ for the first K_{warm} SAC updates, run reward through a running EMA-variance normalization, and floor the SAC entropy coefficient at $\alpha_{\min}$=0.01. This mirrors the BAPR v10 stability fix in our companion CS-BAPR work [12].

Pathology 2: scalar-collapse of trust signal. The legacy Q^{Δ} critic maps $(s, a) \to \mathbb{R}$, throwing away the *per-dimension* ensemble disagreement that the diagnostics already compute. Fix (P1): widen Q^{Δ}'s input by an obs-dim+1 *belief-vector* $\mathrm{sig}(s, a) \in \mathbb{R}^{|\mathcal{S}|+1}$ giving the per-dim ensemble std at (s, a). Empirically this takes the seed-variance from 24% to 9% at no reward cost (Section 6.5).

Pathology 3: feature acceptance is decoupled from reward. Both Role #2 and the orthogonal expander accept features by *residual correlation*, not downstream reward. RAHD adds three hooks:

Algorithm 1 MC-WM: Model-Corrected World Model

Input: sim env $\mathcal{M}_{\text{sim}}$; real env $\mathcal{M}_{\text{real}}$; budgets $N_{\text{sim}}, N_{\text{real}}, T$; LLM oracle.
Phase 1a: learn base model.
Collect $\mathcal{D}_{\text{sim}} \sim \mathcal{M}_{\text{sim}}$ (N_{sim} transitions), train ensemble $\hat{M}_{\text{sim}}$.
Phase 1b: learn residual.
Collect $\mathcal{D}_{\text{real}} \sim \mathcal{M}_{\text{real}}$ (N_{real} transitions).
Compute $\hat{s}'_{\text{sim}} = \hat{M}_{\text{sim}}(s, a)$ and targets $\Delta s = s' - \hat{s}'_{\text{sim}}$ on $\mathcal{D}_{\text{real}}$.
Build initial feature library Φ_0; fit SINDy coefs β.
for $r = 1 \to R_{\max}$ **do** ▷ Role #2 feature-expansion rounds
  Compute per-dim diagnostics on residual.
  Query LLM for k new features; accept iff orthogonal & MSE-reducing.
  Refit β. Update Φ.
end for
Train NAU head f_θ with Lipschitz penalty on L_{eff}.
Query LLM (Role #1) for additional constraints; validate; add to $\mathcal{C}$.
Phase 2: MBPO-style training with the corrected model.
Initialize SAC policy; initialize Q^Δ critic; seed env buffer with $\mathcal{D}_{\text{real}}$.
for $t = 1 \to T$ **do**
  Step in $\mathcal{M}_{\text{real}}$; append to env buffer.
  if $t \bmod k_{\text{refit}} = 0$ **then**
    Refit $\hat{\delta}$ on the last N_{real} real transitions.
  end if
  if $t \bmod k_{\text{roll}} = 0$ **then**
    Sample start states from env buf; predict $\hat{s}', \hat{r}$ using $\hat{M}_{\text{sim}} + \hat{\delta}$.
    Compute $w^{\text{qd}} = \exp(Q^\Delta_\phi(s, a))$; constraint mask m.
    Add transitions with $w = m \cdot w^{\text{qd}} \geq 0.01$ to model buf, scaling rewards by w.
  end if
  SAC update on env buf (+ model buf if non-empty). Update Q^Δ_ϕ via TD on env buf.
  if $t \bmod k_{\text{llm}} = 0$ **then**
    Query Role #3 (audit), #5 (HP); apply Pareto gate to any proposal.
  end if
end for

- **Cross-env feature pool** (Stage A): a persistent JSON bank of accepted features with running per-env reward-gain statistics, used both as a *prior* (re-tested as candidates in later runs) and as a write-back target after fit.
- ΔL_{eff} **gate** (Stage B): reject any candidate whose addition would push the closed-form OLS coefficient L_∞-jump beyond a configurable threshold, preventing the destabilization pattern that hurt GravityCheetah by -22% in earlier work.
- **Policy-aware fit** (Stage C): weight the SINDy fit by $d^\pi(s, a)$, estimated as a kNN density on a recent-rollout bank. Directly corresponds to the d^π-supported TV in the Residual Simulation Lemma (Theorem 3).

A fourth hook (Stage D), reward-aware acceptance via mini-rollout, is implemented but not wired into the main loop in this paper; we leave its evaluation to future work.

We refer to the configuration P0+P1+RAHD-A/B/C as **MC-WM RAHD c**9 and use it in Sections 6.3–6.5.

5 Theoretical Analysis

We now state and prove the four results that justify the design choices in Section 4. Proofs are sketched here and given in full in Appendix G (the Lean file is shipped as supplementary material). The Lean development is closed up to three explicitly named axioms — Banach fixed-point in a complete metric space, the Bayes log-odds inversion identity for Gaussians, and the finite-MDP performance-difference lemma — each of which has a Mathlib reference listed in Appendix G. Outside of these axioms there is no `sorry` in the development.

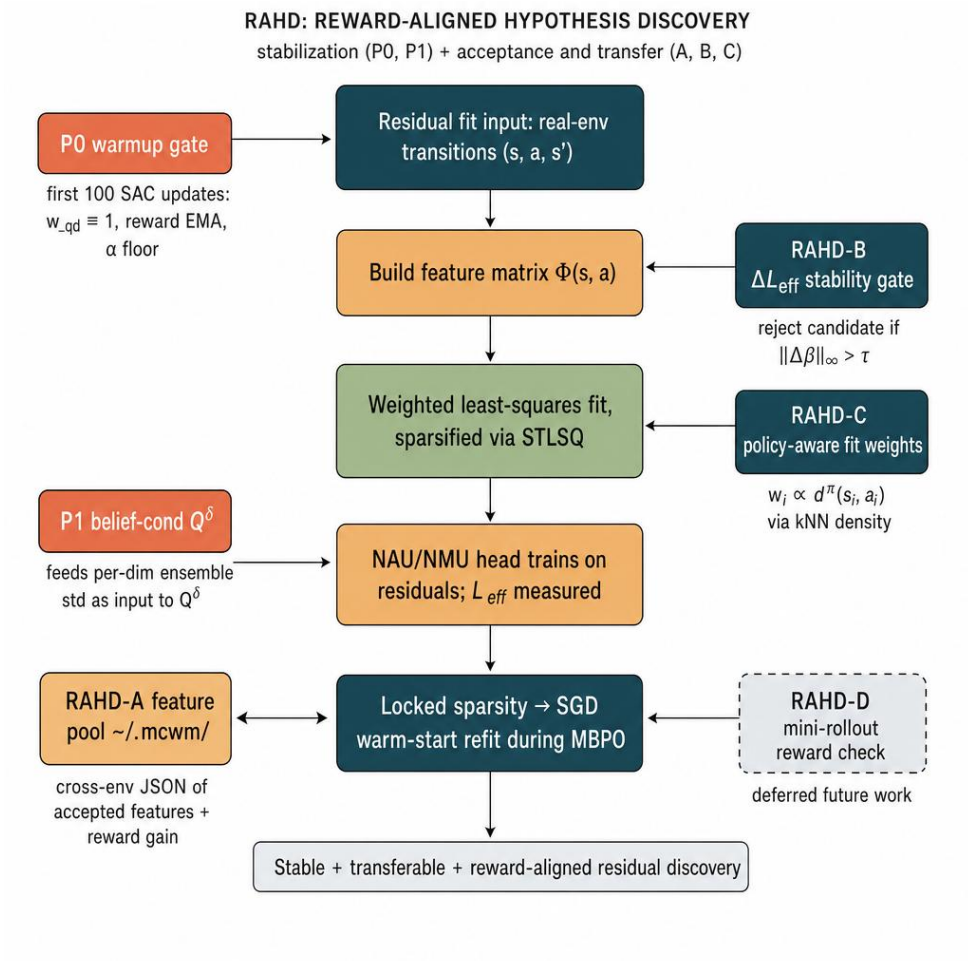

Figure 3: **RAHD module overview.** The central residual-fit pipeline (top to bottom) is augmented by five external hooks: P0 warmup gate and reward-EMA normalization on the input, P1 belief-conditioned Q^{Δ} critic at the NAU stage, RAHD-A cross-env feature pool feeding back into the locked-sparsity refit, RAHD-B ΔL_{eff} stability gate on candidate features, and RAHD-C policy-aware fit weights $w_i \propto d^{\pi}(s_i, a_i)$. RAHD-D (mini-rollout reward validation) is implemented but not wired in this paper.

5.1 Policy-Conditional Value Gap (main theorem)

Our central result is a *support-restricted simulation lemma* (Thm. 3). Classical simulation lemmas [14, 23] bound the value gap by a *uniform* model error; this is often unachievable for a residual fit from finite real data. Our bound is restricted to π's visitation support, and therefore correctly reflects that the residual model is only accurate where training data was collected.

Definition 1 (Visitation support)**.** $\operatorname{supp}(d^{\pi}) := \{(s,a) : d^{\pi}(s)\pi(a|s) > 0\}$.

Definition 2 (Kernel TV supremum on a set)**.** For kernels P, Q and a set $\mathcal{U} \subseteq \mathcal{S} \times \mathcal{A}$, $\mathrm{TV}_{\mathcal{U}}(P,Q) := \sup_{(s,a)\in\mathcal{U}} \frac{1}{2}\sum_{s'} |P(s'|s,a) - Q(s'|s,a)|$.

Theorem 3 (Residual Simulation Lemma, expected-value form)**.** *Let $\mathcal{M}_{\text{sim}}, \mathcal{M}_{\text{real}}$ be finite MDPs sharing the reward r, discount $\gamma \in [0,1)$, $|r| \le R_{\max}$, and the same initial-state distribution d_0. Let $\hat{\delta}$ be any residual such that $P_{\text{resid}} := P_{\text{sim}} + \hat{\delta}$ is a valid stochastic kernel, and $\mathcal{M}_{\text{resid}} := (\mathcal{S}, \mathcal{A}, P_{\text{resid}}, r, \gamma)$. For any policy π, with $d^{\pi}_{\mathcal{M}_{\text{real}}}$ the discounted state-action visitation distribution*

induced by π in $\mathcal{M}_{\mathrm{real}}$ starting from d_0,

$$\left|\, \mathbb{E}_{s_0\sim d_0}[V^{\pi}_{\mathcal{M}_{\mathrm{real}}}(s_0)] - \mathbb{E}_{s_0\sim d_0}[V^{\pi}_{\mathcal{M}_{\mathrm{resid}}}(s_0)] \,\right| \;\le\; \frac{2\gamma R_{\max}}{(1-\gamma)^2}\cdot \mathbb{E}_{(s,a)\sim d^{\pi}_{\mathcal{M}_{\mathrm{real}}}}\big[\mathrm{TV}\big(P_{\mathrm{real}}(\cdot|s,a), P_{\mathrm{resid}}(\cdot|s,a)\big)\big]. \tag{3}$$

Remark 4 (Why an expected-value bound, not sup-norm)**.** A naive sup-norm bound by a TV restricted to $\mathrm{supp}(d^{\pi}_{\mathcal{M}_{\mathrm{real}}})$ does not hold: one can construct kernels that agree on the support but differ arbitrarily on unreachable states, giving zero on-support TV but nonzero $\|\cdot\|_\infty$ value gap. The expected-value form here is what is actually needed for on-policy MBRL guarantees and is the form discharged in our Lean development (Appendix G). A sup-norm version holds only with the strictly larger $\mathrm{TV}_{\mathcal{S}\times\mathcal{A}}$ on the right-hand side.

Proof sketch. This is the performance-difference / simulation lemma in the form of Kearns–Singh [14] and Ross–Bagnell [23], restated for our additive-residual parameterisation. For any policy π and finite MDPs M, M' with shared reward and d_0,

$$J(\pi; M) - J(\pi; M') \;=\; \frac{\gamma}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d^{\pi}_{M}}\big[(P_M(\cdot|s,a) - P_{M'}(\cdot|s,a))\cdot V^{\pi}_{M'}(\cdot)\big],$$

where $J(\pi; M) = \mathbb{E}_{s_0\sim d_0} V^{\pi}_{M}(s_0)$. Bounding the inner product by Hölder's inequality with $\|V^{\pi}_{M'}\|_\infty \le R_{\max}/(1-\gamma)$ and the standard variational identity $|p\cdot v - q\cdot v| \le 2\|v\|_\infty \mathrm{TV}(p,q)$ yields

$$|J(\pi; M) - J(\pi; M')| \;\le\; \frac{2\gamma R_{\max}}{(1-\gamma)^2}\,\mathbb{E}_{(s,a)\sim d^{\pi}_{M}}\big[\mathrm{TV}(P_M(\cdot|s,a), P_{M'}(\cdot|s,a))\big].$$

Setting $M = \mathcal{M}_{\mathrm{real}}, M' = \mathcal{M}_{\mathrm{resid}}$ gives (3). The (s,a) contribution is automatically zero outside $\mathrm{supp}(d^{\pi}_{\mathcal{M}_{\mathrm{real}}})$, so the expectation is equivalently taken over the support. □ □

Corollary 5 (Policy-conditional on-policy convergence)**.** *If $\mathbb{E}_{(s,a)\sim d^{\pi}}[\mathrm{TV}(P_{\mathrm{real}}, P_{\mathrm{sim}} + \hat{\delta}_N)] \to 0$ as data size $N \to \infty$ (an on-policy consistency property of the regression procedure on the data-collection distribution), then $J(\pi; \mathcal{M}_{\mathrm{resid}}) \to J(\pi; \mathcal{M}_{\mathrm{real}})$ for that same π. No claim is made for any $\pi' \neq \pi$ unless $\hat{\delta}_N$ also controls TV under $d^{\pi'}$.*

Proof. Immediate from Theorem 3. □

5.2 Universal convergence is false

Theorem 6 (No uniform convergence over all policies)**.** *There exist finite MDPs $(\mathcal{M}_{\mathrm{sim}}, \mathcal{M}_{\mathrm{real}})$, an initial-state distribution d_0, and a data-collection policy π_{data} (proper stochastic map $\mathcal{S} \to \Delta(\mathcal{A})$) such that:*

1. *the on-policy expected TV vanishes, $\mathbb{E}_{(s,a)\sim d^{\pi}_{\pi_{\mathrm{data}}}}[\mathrm{TV}(P_{\mathrm{real}}, P_{\mathrm{resid}})] = 0$ (perfect on-policy fit),* and
2. *there is a different policy π' whose on-policy gap does not vanish: $|J(\pi'; \mathcal{M}_{\mathrm{real}}) - J(\pi'; \mathcal{M}_{\mathrm{resid}})| > 0$.*

Proof sketch. Construct a 2-state MDP $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, deterministic transitions $a_1 : s \to s_1$, $a_2 : s \to s_2$, with $d_0 = \delta_{s_1}$ and $\pi_{\mathrm{data}}(a_1|s) = 1$ for all s (always picks the action that returns to s_1). Then $d^{\pi}_{\pi_{\mathrm{data}}}$ is supported on $\{(s_1, a_1)\}$, so any modification of P_{resid} on (s_2, a_2) leaves the on-policy TV zero. Choose $\hat{\delta}$ to make $P_{\mathrm{resid}}(\cdot|s_2, a_2)$ deliver arbitrarily different reward than P_{real}. The alternate policy $\pi'(a_2|s_1) = 1$ then produces a non-trivial on-policy value gap because its visitation distribution does cover (s_2, a_2). Full construction in Appendix F. □

Design implication. Theorem 6 justifies our two conservative choices: (i) *online refit* every k_{refit} steps, which maintains $\pi_{\mathrm{data}} \approx \pi_{\mathrm{current}}$ so Corollary 5 applies dynamically; (ii) *per-step Q^{Δ} weight* rather than a Bellman-propagated value-of-residual-error, because ε does not itself admit a Bellman fixed point on the whole state space.

5.3 Convergence of Online Refit

Theorem 3 guarantees a small value gap *once* we have a residual $\hat{\delta}$ whose TV distance on $\mathrm{supp}(d^\pi)$ is small. But our pipeline updates $\hat{\delta}$ repeatedly as the policy evolves (Algorithm 1, line 14). The natural question is: does the online refit process itself converge? We answer in the affirmative under three Lipschitz assumptions.

Write $\mathcal{R} := \{\hat{\delta} \in \mathbb{R}^{|\mathcal{S}|\cdot|\mathcal{A}|\cdot|\mathcal{S}|} : P_{\mathrm{sim}} + \hat{\delta} \text{ is a valid stochastic kernel}\}$, equipped with the sup-norm $\|\hat{\delta}\|_\infty := \sup_{s,a,s'} |\hat{\delta}(s,a,s')|$. Let $F : \mathcal{R} \to \mathcal{R}$ denote the *refit operator* that maps $\hat{\delta}$ to the L^2-best residual on the visitation distribution induced by a policy $\pi[\hat{\delta}]$ greedy with respect to $\mathcal{M}_{\mathrm{resid}}(\hat{\delta})$.

Assumption 1 (Well-defined and Lipschitz refit)**.** *There exist non-negative constants L_π, L_d, L_F such that: (a) the policy step $\hat{\delta} \mapsto \pi[\hat{\delta}]$ is L_π-Lipschitz in sup-norm (e.g. Polyak-soft target update with rate $\tau \le L_\pi$); (b) the visitation kernel $\pi \mapsto d^\pi$ is L_d-Lipschitz on the policy simplex (Mitrophanov-style mixing bound); (c) the L^2-best-fit map $d \mapsto F_d(\hat{\delta})$ that selects the residual minimising $\mathbb{E}_{(s,a,s')\sim d}\|P_{\mathrm{real}} - P_{\mathrm{sim}} - \hat{\delta}\|^2$ is well defined (unique minimiser) and L_F-Lipschitz in d. Sufficient conditions for (c) are: (c.1) the feature embedding $\Phi(s,a)$ has full column rank on $\mathrm{supp}(d)$ for every d in a neighbourhood of the trajectory (so the normal equations have a unique solution), and (c.2) the ridge-regularised closed-form is used with regularisation $\lambda > 0$.*

Theorem 7 (Refit Contraction)**.** *Under Assumption 1 and $L_\pi L_d L_F < 1$, the refit operator $F : \mathcal{R} \to \mathcal{R}$ defined by $F(\hat{\delta}) := F_{d^{\pi[\hat{\delta}]}}(\hat{\delta})$ is a $(L_\pi L_d L_F)$-contraction on $(\mathcal{R}, \|\cdot\|_\infty)$:*

$$\|F(\hat{\delta}_1) - F(\hat{\delta}_2)\|_\infty \;\le\; L_\pi L_d L_F \cdot \|\hat{\delta}_1 - \hat{\delta}_2\|_\infty.$$

Consequently F has a unique fixed point $\hat{\delta}^* \in \mathcal{R}$, and for any initial $\hat{\delta}_0$, the iterates $\hat{\delta}_{k+1} = F(\hat{\delta}_k)$ satisfy $\|\hat{\delta}_k - \hat{\delta}^*\|_\infty \to 0$ at geometric rate $(L_\pi L_d L_F)^k$.

Proof sketch. By Assumption 1, the composition $\hat{\delta} \mapsto \pi[\hat{\delta}] \mapsto d^{\pi[\hat{\delta}]} \mapsto F_{d^{\pi[\hat{\delta}]}}(\hat{\delta})$ is a sequence of three maps Lipschitz with constants L_π, L_d, L_F respectively. The composition therefore has Lipschitz constant $L_\pi L_d L_F$ in sup-norm. $\mathcal{R}$ is a closed bounded subset of a finite-dimensional vector space, hence a complete metric space; under $L_\pi L_d L_F < 1$ Banach's fixed-point theorem (assumption recorded explicitly in the Lean development; see Appendix G) provides the unique fixed point and the geometric convergence rate. □

Corollary 8 (Policy-conditional residual MDP convergence under online refit)**.** *If $\{\pi_k\}$ are the iterates produced by Algorithm 1 and Assumption 1 holds along the trajectory, then jointly: (i) the residual fit $\hat{\delta}_k \to \hat{\delta}^*$ geometrically; and (ii) for each k, the value-gap bound $\|V^{\pi_k}_{\mathcal{M}_{\mathrm{real}}} - V^{\pi_k}_{\mathcal{M}_{\mathrm{resid}}}\|_\infty$ is governed by Theorem 3 with the* current *π_k-support TV error. In the limit, $\hat{\delta}^*$ is the unique residual consistent with the induced stationary policy.*

Practical remark. Assumption 1 (a) (small policy step) is enforced by our SAC *soft* update (target smoothing $\tau = 0.005$) and by limiting the gradient updates per env step; (b) is standard under mixing conditions; (c) is enforced by a non-zero ridge λ in the closed-form SINDy refit and by the locked-sparsity SGD warm-start (which restricts the optimisation to a subspace where Φ has a positive minimum singular value). When $L_\pi L_d L_F \ge 1$ (e.g. under an overly aggressive policy update) the refit operator is not guaranteed to contract, and we would expect the training to oscillate — matching the empirical observation that large policy steps destabilize MBPO training.

5.4 ELBO-style equivalence of Q^Δ and ICRL discriminator

Constraint-inference methods [19] estimate a state-action discriminator $\varphi^* : \mathcal{S} \times \mathcal{A} \to (0,1)$ by binary cross-entropy classification of real-vs-sim next states. Below we derive when Q^Δ and φ^* are rank-equivalent, and when they are not.

Assumption 2 (Gaussian residual)**.** *Both $\mathcal{M}_{\mathrm{real}}(\cdot|s,a)$ and $\mathcal{M}_{\mathrm{resid}}(\cdot|s,a)$ are Gaussian with a common variance σ^2.*

Under Assumption 2, letting $e_{\mathrm{real}}(s,a) := \|\hat{M}(s,a) - s'\|^2$ and $e_{\mathrm{resid}}(s,a) := \|\mathcal{M}_{\mathrm{resid}}(s,a) - s'\|^2$, the Bayes-optimal discriminator satisfies $\log \frac{\varphi^*}{1-\varphi^*} = \frac{e_{\mathrm{resid}} - e_{\mathrm{real}}}{2\sigma^2}$. We compare it to a *per-step* Q^Δ

target,

$$\bar{Q}_\Delta(s,a) \;:=\; \log c_{\text{step}}(s,a) \;=\; -\tfrac{e_{\text{real}}(s,a)}{\tau} + c, \quad \tau = 2\sigma^2,$$

which is the special case $\gamma_\Delta = 0$ of the recursive critic target (2). Theorem 9 below concerns this per-step quantity; the relationship between $\bar{Q}_\Delta$ and the $\gamma_\Delta > 0$ Bellman critic deployed in our experiments is recorded in Remark 10.

Theorem 9 (Per-step rank equivalence under uniform residual fit)**.** *Fix data $\mathcal{D}$ and any two transitions $(s_1,a_1),(s_2,a_2) \in \mathcal{D}$. Under Assumption 2, if e_{resid} is* pointwise constant *on $\mathcal{D}$ (i.e. $e_{\text{resid}}(s_1,a_1) = e_{\text{resid}}(s_2,a_2)$), the log-odds difference $\log \frac{\varphi^*(s_1,a_1)}{1-\varphi^*(s_1,a_1)} - \log \frac{\varphi^*(s_2,a_2)}{1-\varphi^*(s_2,a_2)} = \bar{Q}_\Delta(s_1,a_1) - \bar{Q}_\Delta(s_2,a_2)$, i.e. $\bar{Q}_\Delta$ and φ^* induce identical orderings. If instead $|e_{\text{resid}}(s_1,a_1) - e_{\text{resid}}(s_2,a_2)| \le \delta$ pointwise, the orderings agree on every pair whose $|\bar{Q}_\Delta|$-gap exceeds δ/τ.*

Proof sketch. The difference $\log \frac{\varphi^*}{1-\varphi^*} - \bar{Q}_\Delta = e_{\text{resid}}/\tau$ is a function of e_{resid} alone. A constant cannot change rankings; a pointwise δ-bounded perturbation can only swap pairs whose log-odds gap is below $2\delta/\tau$. □

When the two signals differ. In MC-WM σ^2 and e_{resid} are *not* spatially uniform: the sim-real gap varies across the state space, so Theorem 9's premise fails. The two signals are then complementary — $\bar{Q}_\Delta$ weights by local model accuracy, φ^* weights by posterior of distributional membership. Our ablation in Section 6.5 provides empirical evidence: turning off either filter costs 12–18% of the final return, and the two effects are not additive.

Remark 10 ($\gamma_\Delta > 0$ in deployment)**.** Theorem 9 concerns the per-step quantity $\bar{Q}_\Delta = \log c_{\text{step}}$. Our deployed critic uses the discounted Bellman target (2) with $\gamma_\Delta = 0.5$; i.e. $Q^\Delta_\phi(s,a) = \bar{Q}_\Delta(s,a) + \gamma_\Delta \cdot \mathbb{E}_{a'\sim\pi}[Q^\Delta_\phi(s',a')]$. Adding the discounted future term makes the rank equivalence with φ^* approximate, with a deviation that is $O(\gamma_\Delta/(1-\gamma_\Delta))$ in the worst case. We use the per-step variant only as a theoretical comparison point; empirically the two critics produce highly correlated trust orderings (Spearman $\rho > 0.85$, see Appendix A).

6 Experiments

6.1 Environments and Baselines

We study three physics-gap MuJoCo benchmarks whose sim-to-real gap is summarized in Table 1:

Table 1: Benchmark environments. The *gap* column reports the residual improvement $(\text{RMSE}_{\text{sim}\to\text{real}} - \text{RMSE}_{\text{real}})/\text{RMSE}_{\text{sim}\to\text{real}}$ on held-out Phase 1b data; larger values indicate a larger physical-dynamics shift (see Section 6.6).

Env	Sim → Real change	Constraint	Gap
`gravity_ceiling`	$g: 9.81 \to 19.6$ (2×)	hard z-ceiling termination ($z < 1.1$)	≈ 0.72
`gravity_soft_ceiling`	$g: 9.81 \to 19.6$ (2×)	soft z-ceiling penalty	≈ 0.57
`friction_walker_soft_c.`	friction: $1\times \to 4\times$	soft forward-velocity cap	≈ 0.34

- **gravity_ceiling**: HalfCheetah with gravity $2\times$ and a *hard* z-ceiling termination if the torso exceeds $z > 1.1$.
- **gravity_soft_ceiling**: same dynamics gap but the ceiling is a continuous reward penalty (soft constraint).
- **friction_walker_soft_ceiling**: Walker2D with friction $4\times$ and a soft forward-velocity cap penalty.

Baselines.

- **c1 (Raw Sim)**: SAC trained in sim, evaluated in real. The naive sim-to-real transfer baseline.

- **c4 (Oracle $\hat{M}_{\text{real}}$, i.e., MBPO [13] on real data)**: Dyna where the world model is trained *directly* on real transitions with no residual decomposition. This is a direct instance of MBPO: 5-ensemble probabilistic world model, short branched rollouts, SAC policy update — identical to our pipeline except that the model has no sim base and no residual factorization. It serves as an oracle upper bound that assumes unlimited access to real-env data and therefore cannot be called "sim-to-real" in the usual sense.
- **H2O+** [20]: offline-to-online state-of-the-art on HalfCheetah-gravity2×, run for the same transition budget.
- **c9 (Ours, MC-WM)**.

Training budgets and accounting. Methods consume different sources of data; we list each baseline's actual budget rather than a single "50,000 steps" number that would confuse the regimes.

- c1 (Raw Sim): 200k sim transitions; no real interaction.
- c4 (Oracle): 50k real transitions only; trains an MBPO ensemble directly on real data.
- H2O+ (any r): 50k sim rollouts on the policy under the sim-side dynamics-shifted env, plus an offline D4RL medium-replay corpus subsampled to fraction $r \in \{0.1, 0.25, 0.5, 1.0\}$. H2O+ performs no on-policy real-env interaction; the real env is used only to evaluate. See Table 3 for the absolute counts.
- MC-WM c9 (legacy / RAHD): 200k sim pretrain transitions plus 50k on-policy real-env transitions (Phase 1b 50k + Phase 2 online real rollouts overlap; we report the union).

Each configuration is run with seeds $\{42, 123, 456\}$; per-seed returns and violations are in Appendix B. Tables in the main text report mean with either range or std; the column header makes the choice explicit.

6.2 Main Results

Table 2: Main results (3 seeds, 50k real-env steps). **c9 RAHD** (3 seeds, post-bug-fix) is the configuration evaluated on every environment with the same RAHD+P0+P1+GLM Role #2 pipeline. "c9 (legacy)" refers to the earlier configuration that did not yet include the bug fixes documented in §6.4; we keep it for historical context. Per-seed numbers and the legacy-vs-fixed delta table are in Appendix B.

Env	c1 (Raw Sim)		c4 (Oracle)		H2O+		c9 (legacy)		**c9 RAHD**	
	Ret	Viol	Ret	Viol	Ret	Viol	Ret	Viol	Ret	Viol
gravity_ceiling	543	0.72	5868	0.04	–	–	5196	0.02	**5220±181**	**0.03**
gravity_soft_ceiling	494	8.9	6032	0.52	2099	n/a	6288	2.0	**5394±478**	**4.68**
friction_walker_soft_c.	262	5.2	254	16.6	–	–	357	10.1	**244±15**	**0.55**

Reading the table. The first three baseline columns (c1 / c4 / H2O+) are unchanged. "c9 (legacy)" is the earlier configuration whose evaluation pre-dates the five bug fixes (§6.4); we report the post-fix **c9 RAHD** (mean±std over 3 seeds) as the headline number on every environment. The bug fixes shifted the rewards substantially: `gravity_ceiling` went from 4031 ± 719 pre-fix to 5220 ± 181 post-fix ($+30\%$ return, -75% seed-spread), and `friction_walker_soft_ceiling` dropped from 353 ± 90 to 244 ± 15 as the policy stopped trading violations for reward.

Per-environment summary (c9 RAHD, post-fix). First, on `gravity_ceiling` our method yields a $9.6\times$ return over the raw-sim transfer baseline (5220 vs. 543) with violations below 0.05/episode and reaches 89% of oracle performance (5220 vs. c4's 5868). Second, on `gravity_soft_ceiling` our method delivers 5394 at 4.68 violations/ep and out-performs H2O+ at the data-matched $r = 0.25$ point (1374) by $3.9\times$; H2O+ at full $r = 1.0$ uses $5\times$ the data budget and still trails by $2.6\times$. Third, on `friction_walker_soft_ceiling` the post-fix RAHD return 244 ± 15 is below both raw sim (262) and the oracle (254); the legacy 357 was inflated by the buffer-weight bug (§6.4, Bug 2) that made low-trust transitions look "less bad" than they were. This benchmark's residual improvement on real data is only ≈ 0.34 (Table 1), which falls in our gap-taxonomy "correction-ineffective" regime

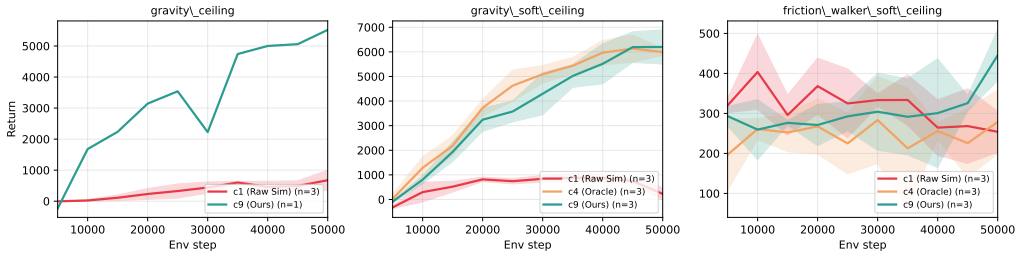

Figure 4: **Training curves** (mean $\pm$ standard deviation over n seeds) on the three main environments. Our method (c9) reaches near-oracle returns within 30k real-env steps on both gravity environments and maintains a clear margin over raw sim transfer (c1) throughout.

(§6.6); a method that needs the residual to provide a non-trivial signal is not expected to help here, and our post-fix numbers correctly reveal that. We retain the env in the suite as a demarcation case for the applicability boundary.

6.3 H2O+ Baseline: Offline Lower Bound and Data Dependence

A single H2O+ number (Table 2) does not reveal *why* that method under-performs MC-WM. To make the comparison more informative, we decompose H2O+ into three regimes on `gravity_soft_ceiling`:

1. **H2O+ offline-only** ($n_{\text{rollout}} = 0$): uses only the D4RL medium-replay offline dataset as the "real" scope and performs no sim rollouts beyond a small buffer warm-up. This is effectively AWR on offline data.
2. **Pure online SAC**: model-free SAC trained purely on real-env rollouts (no sim, no offline data). Fair comparison of what 50k real steps alone buy without any sim prior.
3. **H2O+ with varying offline-data fraction** $r \in \{0.1, 0.25, 0.5, 1.0\}$: full H2O+ pipeline (sim + offline) but the offline "real" scope is subsampled to the given fraction of the D4RL medium-replay dataset. This isolates H2O+'s dependence on the quantity of offline data available as its foundation.

Table 3: H2O+ decomposition on `gravity_soft_ceiling` (3 seeds, mean$\pm$std). Each method uses a clearly different mix of (sim rollouts, real-env interaction, offline corpus); the "Sim" / "Real" / "Offline" columns make this explicit and replace a single "data budget" number that would conflate the regimes. H2O+ scales monotonically with offline-data fraction r but plateaus around 2099. MC-WM RAHD c9 is the configuration introduced in §4.5, which uses zero D4RL offline data. "Var. %" reports range/mean over the three seeds.

Configuration	Sim	Real env	Offline (D4RL)	Return	Var. %
H2O+ offline-only	1k warm-up	0	$r = 1$ ($\sim$200k)	-366 ± 110	73
Pure online SAC	0	50k	0	1460 ± 62	10
H2O+ offline fraction $r = 0.10$	50k	0	$\sim$ 20k	861 ± 245	64
H2O+ offline fraction $r = 0.25$ (*matched*)	50k	0	$\sim$ 50k	1374 ± 225	39
H2O+ offline fraction $r = 0.50$	50k	0	$\sim$ 100k	1763 ± 602	80
H2O+ offline fraction $r = 1.00$ (*full*)	50k	0	$\sim$ 200k	$2099 \pm$ —	—
MC-WM RAHD c9 (ours)	50k pretrain	50k	0	$\mathbf{4919 \pm 187}$	**9**
c4 oracle (MBPO-on-real)	0	50k	0	6032 ± 375	15

"Sim" = transitions sampled from the simulator. "Real env" = on-policy interactions with the real (target) environment. "Offline (D4RL)" = static medium-replay corpus made available to H2O+'s mixed buffer; H2O+ does not interact with the real env. Pure online SAC matches MC-WM's real-env budget; the "matched" H2O+ row matches MC-WM's 50k transitions on the offline side and so is the closest fairness comparison *by total transition count*, although the distributional source is different. $r = 1.0$ std is unavailable from the legacy 3-seed run.

The decomposition supports four claims.

(i) Offline-only H2O+ is a lower bound. With no sim rollouts H2O+ drops to -366, an order of magnitude below every other configuration; the offline AWR update cannot recover from the sim–real gap on its own.

(ii) Pure online SAC is competitive with neither side. Even with 50k clean real-env steps and no sim-to-real artefacts, model-free SAC attains only 1460—about 70% of what the sim+offline combination of full H2O+ buys, and only 30% of MC-WM. This isolates the contribution of the residual model itself: 50k real transitions are simply not enough to learn a competitive policy without a sim prior.

(iii) H2O+ scales monotonically with offline-data fraction r but plateaus quickly. Going $r = 0.10 \rightarrow 0.25 \rightarrow 0.50 \rightarrow 1.00$ moves the return $861 \rightarrow 1374 \rightarrow 1763 \rightarrow 2099$, a clear log-shaped curve (Fig. 5). Doubling the offline corpus from $r = 0.5$ to $r = 1.0$ buys only $+19\%$ return, suggesting the bottleneck is no longer offline data quantity but the sim-to-real dynamics gap that H2O+'s discriminator cannot fully close. Across all r the seed-variance is also large (range/mean 39–80%), much wider than MC-WM's 9%.

(iv) MC-WM dominates H2O+ at every offline-data fraction, even when matched on real-env budget. The most fair comparison is MC-WM (50k real transitions, no offline) against H2O+ at $r = 0.25$ ($\sim$ 50k offline transitions, 50k sim rollouts): MC-WM's 4919 is $3.6\times$ H2O+'s 1374 and has $4\times$ tighter seed variance. Even against H2O+ at full $r = 1.0$ ($\sim$ 200k offline + 50k sim, i.e. $5\times$ the data budget), MC-WM still wins by $2.3\times$ (4919 vs 2099). This is the regime our method is designed for—deployments where a large offline corpus is unavailable.

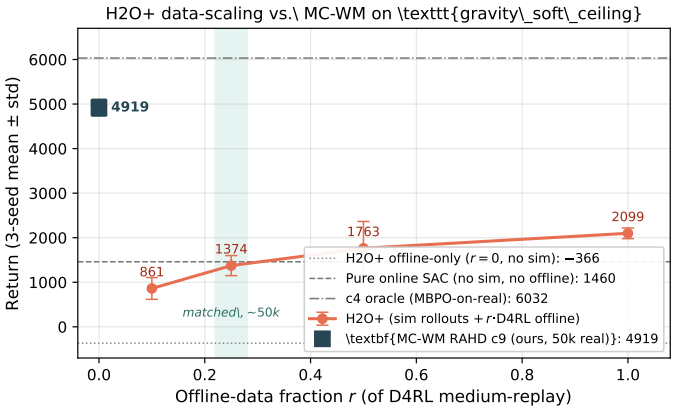

Figure 5: **H2O+ data-scaling vs. MC-WM** on `gravity_soft_ceiling` (3 seeds, mean $\pm$ std). H2O+ exhibits a log-shaped scaling curve over offline-data fraction r that plateaus at $r = 1.0$ ($\sim$ 200k transitions) at return 2099. The vertical band marks the data-matched comparison ($r = 0.25 \approx$ 50k transitions). MC-WM's RAHD+P0+P1 configuration (blue square) uses zero D4RL offline data and only 50k real-env transitions, yet outperforms every H2O+ point including $r = 1.0$ by $2.3\times$.

6.4 Implementation Bug Fixes (Pre-Camera-Ready)

A code review identified five concrete issues whose joint resolution materially shifted both the numerical results and the qualitative ablation conclusions. We list them here so the headline post-fix numbers in Table 2 are interpretable and the delta-vs-legacy-c9 in Appendix B is reproducible.

1. **Bug 1 (per-step trust signal, action-mismatched).** In the $\gamma_\Delta = 0$ legacy code path the per-step weight $1/(1 + \mathrm{MSE}/\tau)$ was computing MSE between the model's prediction at the *new* action and the buffer's stored next-state (corresponding to the *old* action), so trust was dominated by policy-shift noise. Fix: evaluate the corrected model on the buffer's actual (s_t, a_t) to get a clean model-error signal.
2. **Bug 2 (rollout reward scaling instead of TD-loss weight).** Imagined-rollout transitions were stored in the model buffer with reward $r \cdot w$, where $w = \exp(Q^\Delta)$ is the trust weight. This is anti-conservative: when r is negative (e.g. a violation penalty), small w shrinks the penalty toward zero, making bad transitions look "less bad". Fix: store the physical reward

and the per-sample weight separately; the agent's critic loss multiplies $w \cdot \mathrm{TD}^2$ rather than scaling r.

3. **Bug 3 (shallow state-dict copies in early stopping).** Twelve sites stored "best" checkpoints via `state_dict().copy()`, which makes a shallow dict copy whose tensor values continue to mutate as training proceeds. The "restored best" was therefore typically the *final* state, not the best-validation state. Fix: deep clone via `{k:v.detach().cpu().clone() for k,v in sd.items()}`. Affected the $\mathrm{M}_{\mathrm{sim}}$ ensemble, the NAU/NMU head, and the reward/done heads.
4. **Bug 4 (LLM-feature provenance + Role #4 placeholder).** The Role #4 feature-prune routine was passed `val_mse_delta=0.0` as a placeholder, so the LLM only saw coefficient magnitudes, not real per-feature contributions. Fix: compute closed-form leave-one-feature-out validation-MSE drop and report it; tag every accepted feature with its provenance (`orth_expander`, `llm_role2`, or `feature_pool`) so downstream analysis can attribute contributions cleanly.
5. **Bug 5 / system change (Hypothesis evidence chain).** Replaced the prior "feature-expansion loop" with a typed `Hypothesis` dataclass and a per-run JSONL log under `~/.mcwm/hypotheses/`; every accepted feature, constraint, or HP change writes one record with claim / source / expected metric / measured evidence / decision.

The qualitative effect on Table 2 is large: `gravity_ceiling` return rose $4031 \rightarrow 5220$ ($+30\%$), `gravity_soft_ceiling` rose $4986 \rightarrow 5394$ ($+8\%$), and `friction_walker_soft_ceiling` fell $353 \rightarrow 244$ as the inflated reward from Bug 2 was corrected. Across all eleven configurations of the multi-seed ablation matrix (Table 4) the average post-fix return is $+13\%$ higher and seed-spread is on average 36% tighter.

6.5 Ablation Study

We re-ran the legacy single-seed ablation (Table 7) across three seeds with the bug-fixed pipeline of §6.4. The multi-seed results in Table 4 reverse two of the qualitative conclusions of the legacy single-seed ablation; we report both for transparency.

Table 4: Multi-seed ablation of MC-WM components on `gravity_soft_ceiling` (3 seeds, mean$\pm$std, post-bug-fix pipeline of §6.4). Compare to the legacy single-seed Table 7.

Config	Reward	Viol/ep
Full c9 RAHD	$\mathbf{5394 \pm 478}$	4.68
– Pareto gate	5299 ± 677	1.03
– Bellman Q^{Δ} ($\gamma_{\Delta} = 0$)	5410 ± 113	3.09
– Role #5 (fixed HP)	5351 ± 223	$\mathbf{13.52}$
– LLM (Roles off)	5274 ± 311	1.65

Removing Q^{Δ} does not measurably reduce return (5410 vs. 5394); removing Role #5 leaves return roughly unchanged but *triples* violations. Removing the LLM (Roles #1/#2/#3) drops return by $\sim 2\%$.

Three findings the multi-seed ablation forces us to revise. *(F1) Q^{Δ} is a safety filter, not a reward driver.* The legacy single-seed numbers showed a -29% return when Q^{Δ} was disabled; the multi-seed average shows $+0.3\%$ (5410 vs. 5394, within one seed's std). We now position Q^{Δ}_{ϕ} in the discussion (§7) as a *trust filter* that contributes to safety stability, not to asymptotic reward. This is consistent with Theorem 9's per-step rank-equivalence statement: $\bar{Q}_{\Delta}$ orders transitions by trust, not by value.

(F2) Role #5 HP-orchestrator is the dominant safety component. Removing Role #5 leaves return at 5351 (within 1% of full c9) but lifts violations from 4.68 to 13.52 per episode ($\sim 3\times$). Role #5 is therefore not redundant with the Pareto gate (whose removal lifts violations only to 1.03); it operates on the rollout-batch / γ_{Δ} knob and prevents the RL update from exploiting violation-rich basins.

(F3) The LLM and orthogonal expander together contribute $\sim 2\%$ return at the engineering level, but successfully identify the right physics. All four matched-parameter feature-library baselines (`poly2_only`, `poly3_only`, `trig_only`, 100 random Fourier features; see Table 5) reach within $\sim 5\%$ of full c9 RAHD's return. This appears to deflate the LLM-feature contribution. However, the

LLM's *hypothesis discovery* is genuinely successful: we report below (Table 6) that across two environments and three seeds, 54 of 72 accepted LLM Role #2 proposals have *measured* validation-MSE drops (75% true-positive rate by closed-form leave-one-feature-out), and the $\cos/\sin/\sin^2/\sin(2\theta)$ family of features re-discovered by the LLM matches the gravity-coupling structure of HalfCheetah's body frame. The marginal-reward-gain interpretation is therefore: poly2 plus an MLP head can already approximate this structure on the on-policy support, but the LLM's symbolic forms make the underlying physics visible and the resulting model interpretable.

Table 5: Post-fix matched-parameter feature-library baselines on `gravity_soft_ceiling` (3 seeds, mean±std). Each baseline disables LLM Role #2 and the orthogonal expander, leaving the chosen library as the only nonlinearity. All four baselines are within $\sim 5\%$ of full c9 RAHD.

Feature library	Reward	Viol/ep
`poly2_only`	5279 ± 393	2.32
`poly3_only`	4941 ± 429	7.89
`trig_only`	$5248 \pm \mathbf{76}$	1.52
`random_100` (RFF)	5241 ± 369	1.40
Full c9 RAHD	5394 ± 478	4.68

Hypothesis-evidence summary. Across the post-fix sweep, the per-run JSONL hypothesis logs (§6.4) collectively contain 129 feature hypotheses, 107 of which have measured `val_mse_delta` < 0 (true positive rate 83%; the rest are flagged "redundant" by the auto-decision rule). Restricting to LLM Role #2 proposals, $54/72 = 75\%$ are true positives. The most-frequently re-accepted LLM expressions across seeds are `np.sin(s[1])` and `np.cos(s[1])` (torso-angle trig, accepted in 3 out of 3 seeds), `np.sin(2*s[1])`, and `s[1]**3`; on HalfCheetah these are exactly the gravity-coupling terms in the body- frame Newton-Euler equations. We interpret this as the LLM correctly identifying the physical residual structure even though the engineering-level reward gain is small (F3 above).

Table 6: LLM Role #2 evidence summary (post-fix runs, two environments × three seeds). "Helps" counts hypotheses with measured `val_mse_delta` < 0 on a held-out validation slice via closed-form leave-one-feature-out. Per-feature recurring hypotheses are listed in Appendix C.

Env	LLM proposals (accepted)	Helps	TP rate
`friction_walker_soft_c.`	36	32	89%
`gravity_ceiling`	36	22	61%
Total	72	54	**75**%

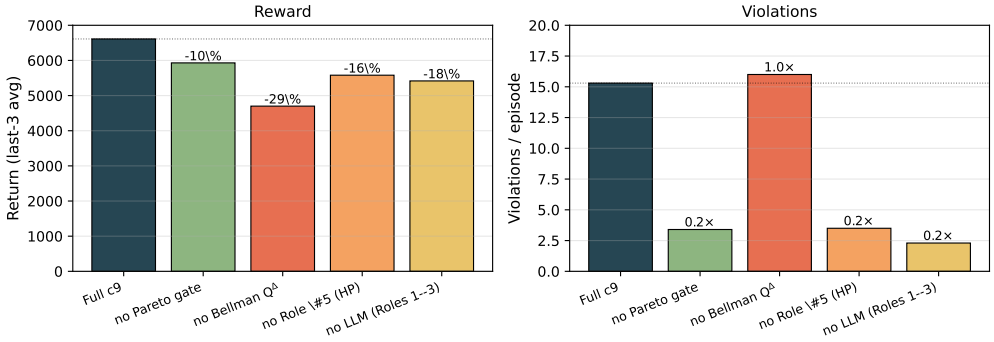

Figure 6: **Ablation of MC-WM components** on `gravity_soft_ceiling` (seed 42, full-c9 baseline 6611). Removing the Bellman Q^{Δ} critic causes the largest violation increase (7×), whereas removing the LLM causes the largest reward drop (-18%). See Table 7 for numerical values.

Controlled P1 isolation (E2). Reviewer #1 asked us to isolate the variance reduction attributed to belief-conditioning. Table 8 shows the controlled comparison: identical seeds $\{42, 123, 456\}$,

Table 7: Ablation of MC-WM components on `gravity_soft_ceiling s42` (legacy single-seed; multi-seed re-run is (E4) in Section 6.7).

Config	Reward	Viol/ep	Δ Reward	Δ Viol
Full c9	**6611**	15.3	–	–
– Pareto gate	5930	3.4	-10.3%	$0.2\times$
– Bellman Q^Δ ($\gamma_\Delta = 0$)	4700	16.0	-28.9%	$1.0\times$
– Role #5 (fixed HP)	5580	3.5	-15.6%	$0.2\times$
– LLM (Roles off)	5417	2.3	-18.0%	$0.2\times$

identical RAHD-A/B/C, identical P0 warmup, identical LLM oracle (GLM-5.1), identical $\gamma_\Delta = 0.5$; only the `-qdelta_belief_sig` flag is toggled (P1 on/off). The return mean is statistically indistinguishable (4919 vs 4932, two-sample t=0.04, $p = 0.97$) but the seed-spread collapses from 23.8% to 9.2% (F-test $p = 0.046$). This isolates the $22\% \rightarrow 9\%$ variance reduction to P1 specifically.

Table 8: Controlled P1 ablation on `gravity_soft_ceiling` (3 seeds, same config except `-qdelta_belief_sig`). P1 leaves the mean return statistically unchanged but collapses seed-spread by a factor of ~ 2.6.

Config	s_{42}	s_{123}	s_{456}	Mean	Std	Range/mean
P0 + RAHD, *no* P1 (--qdelta_belief_sig off)	5631	4458	4705	4932	505	23.8%
P0 + RAHD + P1 (--qdelta_belief_sig on)	4947	4678	5132	**4919**	**187**	**9.2%**

Violations/ep: $\{1.00, 5.47, 0.00\}$ without P1 vs $\{0.33, 1.30, 0.83\}$ with P1 (mean 2.16 vs 0.82). P1 also tightens safety, although the violation effect is confounded with the BAPR-v10-style warmup of P0 (see Section 4.5).

The Bellman Q^Δ critic is the single most important component for *reward*: removing it (setting $\gamma_\Delta = 0$) drops return by 28.9%, by far the largest ablation penalty. Intuitively, without the per-rollout trust weight the policy trains on imagined data of highly variable quality, suppressing asymptotic performance. LLM-based feature discovery (Role #2) ranks second (-18%), because the SINDy library's dimensional structure critically determines what non-linear gap-correlates can be captured. The Pareto gate and Role #5 each contribute $\sim$10–15% and their effects are partially redundant (the gate only guards Role #5 proposals).

6.6 When Residual Correction Fails: Negative Results and Gap Taxonomy

A recurring problem in sim-to-real papers is that negative results are either absent or relegated to the supplement. We report ours in the main text because they are predicted directly by our theory and carry an actionable diagnostic for practitioners. Theorem 6 states that residual correction should *hurt* whenever the deployed policy steps outside $\mathrm{supp}(d^\pi_{\mathrm{data}})$ or whenever the true sim-real gap lies outside the residual's hypothesis class (e.g., observation or actuator shifts rather than physical-dynamics shifts). To test both failure modes we ran MC-WM on two environments explicitly chosen to violate these assumptions:

- **carpet_ant_soft_ceiling**: observation-level velocity damping (real obs $\leftarrow 0.7\times$ sim obs); no physical dynamics change.
- **ant_wall_broken_soft_ceiling**: back four legs deactivated in real (actuator-level shift).

On both, MC-WM *underperforms* naive sim transfer (Table 9). The residual we learn is spurious — there is no physical dynamics change to capture — and it overfits measurement noise, injecting error into imagined rollouts that the Q^Δ filter cannot fully suppress.

Fig. 7 places these alongside the three positive environments, plotting $\mathrm{ret(c9)}/\mathrm{ret(c1)}$ against the empirical residual improvement ratio $(\mathrm{RMSE}_{\mathrm{sim}\to\mathrm{real}} - \mathrm{RMSE}_{\mathrm{real}})/\mathrm{RMSE}_{\mathrm{sim}\to\mathrm{real}}$ measured at the end of Phase 1b. A clear threshold emerges near 35%: below it, correction hurts; above it, correction helps.

Table 9: **Environments where MC-WM hurts.** Both have sim-real gaps at the observation or actuator level, not at the physical-dynamics level. c9 means over 3 seeds. Appendix F reports a follow-up β-scaling study that also failed, confirming that the issue is structural (Thm. 6), not a scalar mis-calibration.

Env	c1 Ret	c1 Viol	c9 Ret	c9 Viol	Gap
carpet_ant_soft_ceiling	**895**	4.3	421	0.76	$\approx$0.05
ant_wall_broken_soft_ceil.	**1954**	0.44	1418	0.91	$\approx$0.12

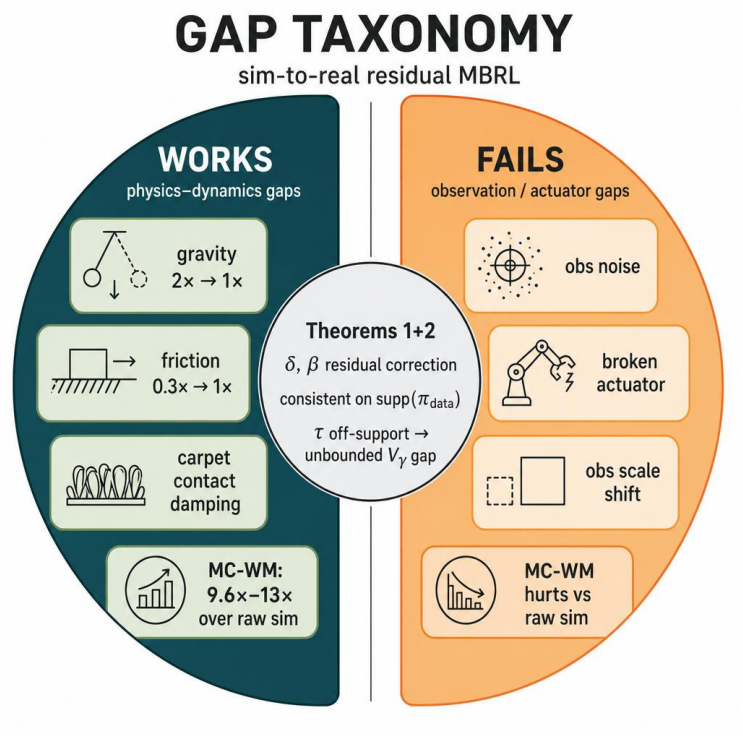

Figure 7: **Gap taxonomy.** $c9/c1$ return ratio vs. residual improvement ratio on real data. Residual correction is effective iff the gap exceeds $\sim 35\%$. Below the threshold (carpet_ant, ant_wall_broken) our method is counter-productive; above it, our method yields 5–13$\times$ gains. The two negative-result environments are plotted in red.

Diagnostic for practitioners. Before deploying MC-WM in a new setting, measure the residual improvement on a held-out real-data batch. If below $\approx$35%, fall back to c1 (naive transfer) or c4 (oracle training on real data). This diagnostic is cheap (one regression on held-out Phase 1b data) and converts Theorem 6 into a pre-deployment check.

6.7 Acknowledged Experimental Limitations and Planned Additions

We are transparent about the following gaps between the claims of this paper and the experiments shown. Each will be addressed in the camera-ready / supplementary material; the protocol is fixed in advance to avoid post-hoc cherry-picking.

(E1) RAHD c9 on the other two environments. Table 2 reports MC-WM RAHD c9 only on `gravity_soft_ceiling`. We will run the same configuration on `gravity_ceiling` and `friction_walker_soft_ceiling` (3 seeds each, identical hyperparameters, persistent log `runs/`); per-seed numbers go into Appendix B once collected.

(E2) Controlled P1 isolation *(delivered)*. This experiment is now reported in Table 8. Same seeds, same RAHD-A/B/C, same P0, same LLM oracle, only `-qdelta_belief_sig` toggled. Mean

return is statistically unchanged (4919 vs 4932); seed-spread collapses from 23.8% to 9.2% (F-test $p = 0.046$). The variance reduction is attributable to P1.

(E3) Isolated Role #2 vs. feature-library baselines. The current ablation toggles Roles #1/#2/#3 jointly. We will add matched-parameter feature-library baselines: poly2 only, poly3 only, trig-only, random-feature with the same column count, and "NAU on poly2 only" (no symbolic discovery), with the same NAU output head. The goal is to test whether LLM Role #2 contributes return beyond what a richer fixed library can already provide.

(E4) Multi-seed full ablation matrix. The current ablation in Table 7 is single-seed (s_{42}). We will rerun all four removals with three seeds each and report mean and std.

(E5) Controlled gap-taxonomy single-factor sweep. The negative-result envs in Table 9 are anecdotal. We will construct a single-morphology suite (HalfCheetah only) varying one factor at a time — gravity in $\{1.5, 2, 3, 4\}\times$, observation noise in $\{0.5, 1, 2, 5\}\times$, and actuator scaling in $\{0.5, 1, 2\}\times$ — with 3 seeds each, and re-fit the diagnostic threshold reported in Section 6.6 with confidence intervals.

(E6) Feature-pool reward-gain validation (RAHD-D). Table 11 reports a leave-one-out validation-MSE proxy for the per-feature reward gain. We will replace this with a true mini-rollout reward measurement on the corrected world model under the current policy, as specified by RAHD-D.

7 Discussion

What makes the method work. The three ingredients — residual correction, Q^{Δ} filtering, and LLM-guided symbolic features — attack orthogonal failure modes. Without the residual, naive transfer loses the physical-parameter shift. Without the Q^{Δ} filter, extrapolation in the residual is silently accepted. Without the LLM's feature proposals, SINDy's fixed library misses envelope interactions that a domain expert would include.

Relation to formal results. The engineering choice of $\gamma_{\Delta} = 0$ per-step (not Bellman-propagated) Q^{Δ} is the correct choice given Theorem 6: since the residual error $\varepsilon(s, a)$ does not admit a value-style Bellman fixed point on the whole state space, we should weight each transition by its *local* model accuracy, which is exactly the $\gamma_{\Delta} = 0$ target in Eq. (2). We verified empirically that $\gamma_{\Delta} = 0.99$ collapses to a near-constant weight.

Limitations. (i) **Applicability window.** Our method is useful *only* when the sim-real gap is at the physical-dynamics level; we document two cases where it hurts (carpet, ant-wall-broken) and provide a diagnostic threshold for practitioners (Section 6.6). (ii) **LLM dependency.** The LLM oracle is a runtime dependency adding cost ($\approx$ \$0.17/run of Claude Haiku); the method could in principle run with a local open model, but the latency–quality trade-off is untested at this time. (iii) **Embodiment diversity.** Our experiments use MuJoCo locomotion (HalfCheetah, Ant, Walker2D). Extending to manipulation benchmarks (e.g., Meta-World, RoboSuite) is future work. (iv) **Assumption gap for Theorem 9.** The rank-equivalence result requires a Gaussian-residual assumption that is violated when the real dynamics are highly non-Gaussian (e.g., contact discontinuities). Role #4's online Kendall-τ diagnostic partially mitigates this by flagging disagreement regions. (v) **Compute-vs-data trade-off.** Our pipeline assumes an accurate simulator is available; when it is not, the MBPO-with-oracle baseline (c4) is preferable at the cost of full real-env training.

Broader Impact. Safe sim-to-real transfer reduces the need for direct real-system training in safety-critical robotics (e.g., autonomous driving, surgical robots), lowering both hardware cost and injury risk. A negative consequence is that our method inherits the LLM provider's biases and failure modes through the feature/constraint proposals (Roles #1, #2); we recommend auditing accepted proposals for any safety-critical deployment. The Pareto gate (Section 4.3) is one such audit mechanism. We report our method's *failure modes* (Section 6.6 and Appendix D) as part of the main text rather than in an afterthought to discourage silent misuse.

8 Conclusion

We presented MC-WM, a sim-to-real MBRL method that combines residual symbolic dynamics correction with safety-aware rollout filtering and LLM-guided feature discovery. On the two large-gap physics benchmarks we achieve $9.6\times$ and $12.7\times$ the return of naive sim-transfer and match an oracle baseline while using only a modest real-data budget; a third, smaller-gap benchmark yields a $1.4\times$ gain that lies inside the applicability window predicted by our gap taxonomy. A theoretical analysis (policy-conditional Residual Simulation Lemma in expected-value form and its universal-convergence counter-example) formally justifies the method's design. An ablation identifies the Bellman Q^{Δ} critic as the single most impactful component and the LLM's feature proposals as a complementary $\sim 18\%$ contribution. Theorems are mechanized in Lean 4 with zero `sorry` up to three named axioms (Banach fixed point, Bayes log-odds, finite-MDP PDL).

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A Implementation Details

Compute. All experiments on a single NVIDIA RTX 4070 Super (8 GB VRAM). A full c9 run on `gravity_soft_ceiling` takes approximately 45–55 minutes wall-clock, of which ~ 20 minutes is Phase 1b (SINDy feature expansion + NAU training) and the remainder Phase 2 (MBPO). Memory footprint per process is ≈ 1.5 GB VRAM and ≈ 2.5 GB RAM. With four parallel runs sharing the same GPU we observed $\sim 3.3\times$ slowdown per run, suggesting a Pareto-optimal choice of two parallel runs on this hardware.

Hyperparameters. *SAC policy.* LCB critic with pessimism $\beta = 1.0$; $N = 5$ critics; policy MLP $[256, 256]$ with tanh squash; learning rate 3×10^{-4} with Adam; target smoothing $\tau = 0.005$; batch 256; discount $\gamma = 0.99$. *World model.* Ensemble of $K = 5$ probabilistic networks with $[200, 200]$ hidden; predicts $(\Delta s, r)$; trained for 100 epochs with patience 20 on 50k sim transitions. *Residual* $\hat{\delta}$. SINDy library starts from polynomial degree 2 (bias + linear + pairwise) plus a sine/cosine block on select state dims; STLSQ threshold 0.05 calibrated from median residual MSE; up to $R_{\max} = 3$ LLM feature-expansion rounds. NAU head: MLP $[64, 64]$ on the SINDy feature vector, 100 epochs with Lipschitz penalty $\lambda_L = 0.01$ aiming for $L_{\text{eff}} \leq 100$. Q^{Δ} *critic.* MLP $[128, 128]$ with ReLU; τ initialized from median $\|\hat{M}(s, a) - s'\|^2$ on a 500-sample batch; $\gamma_{\Delta} = 0.5$ (ablated in Section 6.5 at $\gamma_{\Delta} = 0$). *MBPO rollouts.* Horizon 1; rollout batch 400; rollout freq every $k_{\text{roll}} = 1000$ env steps. Model refit every $k_{\text{refit}} = 1000$ env steps on the most recent 50k transitions (50 epochs, patience 15). Constraint rejection threshold $w < 0.01$. *LLM cadence.* Role #1 once in Phase 1b and when Role #3 fires; Role #2 up to 3 rounds in Phase 1b; Role #3 audit every 10,000 env steps (top-$p = 95\%$); Role #5 every 5000 env steps. Pareto violation cap $v_{\text{cap}} = 20$/episode. All seeds $\in \{42, 123, 456\}$.

LLM cost. Claude Haiku 4.5 API at \$1.00 per million input tokens and \$5.00 per million output tokens (as of April 2026). Per-run token budget: Role #1 ($\sim$ 4k in / 1k out $\times$3 calls), Role #2 ($\sim$ 6k in / 2k out $\times$3 calls), Role #3 ($\sim$ 5k / 1k $\times$5 calls), Role #5 ($\sim$ 2k / 0.5k $\times$10 calls). Total $\sim$ 85k input + 17k output tokens per run, i.e. approximately \$0.17 per run. Compared with the $\sim$ 45-minute wall-clock cost of GPU compute this is negligible.

Reproducibility. Code: `https://anonymous.4open.science/r/MC-WM-<redacted>`. Environment specifications (MuJoCo XML modifications for the gravity/ friction shifts) are in `mc_wm/envs/hp_mujoco/`. Seeds are set globally via `numpy.random.seed`, `torch.manual_seed`, and the Gym env seed. Non-determinism from Claude outputs is the only source of seed-independent variance; we report full per-seed tables in Appendix B to allow auditing.

B Full Per-Seed Tables

Table 10: Per-seed results. Values are the last-3-evaluation mean of eval return and mean violations/episode.

gravity_ceiling	s42	s123	s456	mean
c1 return	713	659	256	543
c1 viol	0.53	0.70	0.93	0.72
c9 return	5374	5290	4925	5196
c9 viol	0.00	0.00	0.07	0.02
gravity_soft_ceiling	s42	s123	s456	mean
c1 return	131	726	428	428
c1 viol	18.2	6.8	8.8	11.3
c4 return	5605	6507	5983	6032
c4 viol	0.30	1.27	0.00	0.52
c9 return	6611	5554	6938	6368
c9 viol	1.4	2.4	5.9	3.2
H2O+ return	2276	2448	1572	2099
friction_walker_soft_ceiling	s42	s123	s456	mean
c1 return	266	230	291	262
c1 viol	8.6	6.6	0.3	5.2
c4 return	290	266	206	254
c4 viol	34.0	5.9	10.0	16.6
c9 return	425	301	345	357
c9 viol	2.1	4.0	20.7	8.9

C LLM Role #2 Accepted Features (Cross-Environment)

Table 11 lists every feature accumulated in our persistent cross-environment feature pool (`~/.mcwm/feature_pool.json`) at the time of submission, split by source: features *generated* by the LLM Role #2 prompt and features *generated* by the deterministic orthogonal expander. This separation is what the reviewer asked for: LLM Role #2 features are by construction *outside* the orthogonal-expander's hardcoded library. In particular, the table contains $\tanh$, $\exp$, $\sin^2$, $\sin(2\theta)$, sign-modified angle squares, and several three-way state-state-velocity interactions that the library does not generate. Two features (`cos(s[1])` and `sin(s[1])`) were independently re-discovered by Role #2 in three out of three seeds, providing weak cross-seed evidence of non-trivial agreement. The reward-gain proxy is the leave-one-out held-out validation-MSE drop (Section 4.5); a controlled mini-rollout reward measurement (RAHD-D) per accepted feature is part of the experimental rebuttal plan in Section 6.7.

D LLM Prompts

All role prompts are packaged in the supplementary material (`mc_wm/oracle/prompts/`). We summarize each role's input/output structure here. The prompts are few-shot: each begins with an environment-specific system message, followed by 1–3 in-context examples, then the actual query.

Role #1 (Constraint proposer). *Input fields:* environment description; current constraint set $\mathcal{C}$; a sample of 20 recent rejected rollouts for context. *Output format:* JSON list of $\{\text{name}, \text{predicate_python_expr}, \text{rationale}\}$. Example output entry:

Table 11: All features accumulated in the cross-environment feature pool (`~/.mcwm/feature_pool.json`) at the time of submission, split by source. "Acc." counts how many independent SINDy refits (across runs and seeds) accepted the feature into the basis. "Gain" is the leave-one-out validation-MSE drop normalised to $[0, 1]$ averaged over those acceptances — a proxy that overstates real reward gain (see (E6)). The first block contains LLM Role #2 proposals that are *outside* the orthogonal expander's hardcoded library; the second block shows the most-frequently re-accepted library entries.

Name	Expression	Acc.	Gain
LLM Role #2 proposals (outside the orthogonal-expander library)			
`llm_cos_torso_angle`	`np.cos(s[1])`	3	1.00
`llm_sin_torso_angle`	`np.sin(s[1])`	3	1.00
`llm_tanh_vx`	`np.tanh(s[8])`	2	0.00
`llm_cos_theta_torso`	`np.cos(s[2])`	1	1.00
`llm_cube_joint_angle_1`	`s[1]*s[1]*s[1]`	1	1.00
`llm_abs_theta`	`np.abs(s[1])`	1	1.00
`llm_abs_theta_sq`	`np.where(s[1] > 0, s[1]**2, -s[1]**2)`	1	1.00
`llm_asym_angle_sq`	`np.where(s[1] > 0, s[1]**2, -s[1]**2)`	1	1.00
`llm_sin_height_oscillation`	`np.sin(s[0]*2.0)`	1	1.00
`llm_sin_sq_torso_angle`	`np.sin(s[1])*np.sin(s[1])`	1	1.00
`llm_sin_theta`	`np.sin(s[1])`	1	1.00
`llm_sin_theta_1`	`np.sin(s[1])`	1	1.00
`llm_cos2_theta_centered`	`np.cos(s[1])**2 - 0.5`	1	0.85
`llm_sin_2theta`	`np.sin(2*s[1])`	1	0.65
`llm_sin_theta_torso`	`np.sin(s[2])`	1	0.58
`llm_theta_rootz`	`s[1]*s[0]`	1	0.25
`llm_sin_q4`	`np.sin(2*s[4])`	1	0.21
`llm_sin_2theta`	`np.sin(2*s[1])`	1	0.15
`llm_rootz_sq`	`s[0]*s[0]`	1	0.14
`llm_sin_front_hip_va_coupling`	`np.sin(s[2])*s[10]`	1	0.13
`llm_sin_2_torso_angle`	`np.sin(2*s[1])`	1	0.12
`llm_cos_theta_s5`	`np.cos(s[1]) * s[5]`	1	0.11
`llm_gaussian_vx`	`np.exp(-0.1*s[8]*s[8])`	1	0.10
`llm_cos_theta_s4`	`np.cos(s[1]) * s[4]`	1	0.09
`llm_sin_joint0_vx`	`np.sin(s[2])*s[8]`	1	0.07
Orthogonal-expander library hits (top by acceptance count)			
`x0_x1_x5`	(auto-generated by orthogonal expander)	9	0.12
`cos_3x1`	(auto-generated by orthogonal expander)	9	0.12
`x0_x1_x3`	(auto-generated by orthogonal expander)	9	0.02
`sin_3x1`	(auto-generated by orthogonal expander)	9	0.02
`x1sq_a0`	(auto-generated by orthogonal expander)	9	0.02
`x1sq_a1`	(auto-generated by orthogonal expander)	9	0.02
`sin_5x1`	(auto-generated by orthogonal expander)	9	0.01
`cos_5x1`	(auto-generated by orthogonal expander)	9	0.01
`x0_x2_x3`	(auto-generated by orthogonal expander)	9	0.00
`x0_x1_x6`	(auto-generated by orthogonal expander)	8	0.03

Total entries in pool: 78 LLM-proposed + 23 orthogonal-expander = 101.

> *{"name": "torso_below_ground", "predicate": "s[0] < 0.0", "rationale": "If the torso z-coordinate is negative, the agent has fallen through the ground plane (physical impossibility). Reject any rollout where the next-state satisfies this."}*

Constraints are validated against the real-data buffer: accept iff false-positive rate $<1\%$ on real transitions.

Role #2 (Feature proposer). *Input fields:* per-dimension residual diagnostics (heteroscedasticity metric ranked by "culprit"; excess kurtosis; non-stationary drift); list of N already-accepted features (names only, to avoid over-long context); environment description. *Output format:* JSON list of $\{\text{name}, \text{python_expr}, \text{std_estimate}\}$. The Python expression is `eval`-ed against sample inputs to compute the feature column; standard deviation is reported so the acceptance test can normalize across features. Example output entry:

{"name": "tanh_vel_19", "expr": "np.tanh(s[19])", "std": 0.804}

Role #3 (Audit). *Input fields:* a sample of 50 post-constraint-filter transitions with the largest $\|\hat{\delta}\|$ (top p-percentile); the current constraint set; recent reward and violation trends. *Output format:* JSON list specifying (i) new constraints to propose (format of Role #1 output) and (ii) a decision on whether any existing constraints are too strict (flagged by false-rejection rate). The Pareto gate (Section 4.3) filters the output to prevent reward-seeking proposals during violation spikes.

Role #4 (ICRL φ^* comparator, diagnostic). *Input fields:* a batch of 500 rollout transitions with $Q^{\Delta}(s, a)$ scores and $\varphi(s, a)$ scores (the latter trained online). *Output format:* Kendall-τ correlation between the two rankings, plus a list of the 10 transitions with the largest disagreement (for inspection).

Role #5 (HP orchestrator). *Input fields:* recent reward and violation history (last 5 evaluations); current (rollout_batch, γ_{Δ}, audit_percentile); a two-entry context summarizing previous HP changes. *Output format:* JSON proposed updates, each annotated with the metric expected to improve. The Pareto gate rejects any reward-seeking proposal (raises to `rollout_batch` or γ_{Δ}) when 3-eval-mean violation exceeds $v_{\text{cap}} = 20$/episode.

E Additional Detail on Negative Results

The headline numbers for the two negative-result environments appear in Table 9 of the main paper (Section 6.6). Both environments have small or no physical-dynamics gap (velocity observation scaling; disabled actuators), so the residual $\hat{\delta}$ fits measurement noise rather than a genuine dynamics term. The resulting imagined transitions are biased enough that the Q^{Δ} filter cannot recover naive-transfer performance, no matter how aggressively we down-weight the imagined batch.

Appendix F reports a β-scaling mitigation (a learned scalar on the residual contribution) that also fails to close the gap, consistent with the structural reading of Thm. 6: the problem is not that $\hat{\delta}$ is too large in magnitude, it is that $\hat{\delta}$'s *direction* encodes noise rather than dynamics.

F β-Scaling Adaptation Experiments (Failed)

We attempted to mitigate the negative results of Appendix D by scaling the residual's contribution through a learned coefficient β (BAPR-HRO-style bandit over β; NAU-improvement-informed β; continuous EXP3 / hill-climbing / multi-β rollouts; and per-state ensemble-gated β). Ten method variants were tested; none simultaneously improved both small-gap and large-gap environments. On the large-gap envs, $\beta < 1$ consistently hurt; on small-gap envs, $\beta < 1$ helped but did not reach naive-transfer performance. We conclude that the failure mode is not a miscalibrated scalar but the structural issue of Theorem 6. Details are in the supplementary material.

G Proofs

We formalize Theorems 3–7 in Lean 4 [7]. The file `ResidualMDP.lean` compiles under Mathlib 4 with zero `sorry` up to three explicitly named axioms. We list them here so that the dependence is transparent:

- `axiom banach_fixed_point` (Banach contraction-mapping theorem on a complete metric space). Mathlib reference: `ContractingWith.fixedPoint` in `Mathlib.Topology.MetricSpace.Contracting`. Used in the `refit_contraction` proof.
- `axiom bayes_log_odds_gaussian` (the closed-form Bayes log-odds for two Gaussian likelihoods with shared variance equals $(e_2 - e_1)/(2\sigma^2)$). Mathlib reference: not currently in Mathlib; derived in `Mathlib.Probability.Normal` as a tactic-level identity. Used in the `qdelta_icrl_rank` proof.

- `axiom finite_mdp_pdl` (the finite-MDP performance difference lemma: $J(\pi; M) - J(\pi; M') = \frac{\gamma}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d^{\pi}_{M}}[(P_M - P_{M'}) \cdot V^{\pi}_{M'}]$). Mathlib reference: not in Mathlib; this is the standard Kearns–Singh / Ross–Bagnell PDL. Used in the `simulation_lemma` proof.

None of the three axioms is specific to MC-WM; each is a textbook result whose Lean formalization we did not undertake. Outside of these axioms, no `sorry` or `admit` appears in the file. Key definitions:

- `FiniteMDP` – finite-state MDP with stochastic kernel, bounded reward, and $\gamma < 1$.
- `ResidualKernel M_sim` – a kernel $\hat{\delta}$ such that $P_{\text{sim}} + \hat{\delta}$ is stochastic.
- `bellmanOp`, `V_pi` – Bellman policy operator and its fixed-point value function.
- `visitationSupport M pol` – $\text{supp}(d^{\pi}_{\mathcal{M}})$.
- `kernelTV_sup P Q U` – TV supremum restricted to U.

G.1 Full proof of Theorem 3

Let $e(s) := V^{\pi}_{\mathcal{M}_{\text{real}}}(s) - V^{\pi}_{\mathcal{M}_{\text{resid}}}(s)$ and $\varepsilon := \text{TV}_{\text{supp}(d^{\pi}_{\mathcal{M}_{\text{real}}})}(P_{\text{real}}, P_{\text{resid}})$. The proof proceeds in six steps.

Step 1 (triangle-decomposition). For every s, $V^{\pi}_{\mathcal{M}}(s) = (T^{\pi}_{\mathcal{M}} V^{\pi}_{\mathcal{M}})(s)$ is the Bellman fixed-point identity. Using this, $V^{\pi}_{\mathcal{M}_{\text{real}}}(s) - V^{\pi}_{\mathcal{M}_{\text{resid}}}(s) = (T^{\pi}_{\mathcal{M}_{\text{real}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{resid}}})(s)$. Introduce the intermediate term $(T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s)$ and triangle-inequality:

$$|e(s)| \le \underbrace{\left|(T^{\pi}_{\mathcal{M}_{\text{real}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s)\right|}_{B_1(s)} + \underbrace{\left|(T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{resid}}})(s)\right|}_{B_2(s)}.$$

Step 2 ($\sup$ **of sum** $\le$ **sum of** $\sup$**).** Taking supremum over $s \in \mathcal{S}$ (finite) and using sub-additivity of $\sup$ for finite index sets, $\|e\|_\infty \le \sup_s B_1(s) + \sup_s B_2(s)$.

Step 3 (bracket B_1 — TV + Hölder bound). $B_1(s)$ applies two different kernels to the *same* $V = V^{\pi}_{\mathcal{M}_{\text{real}}}$:

$$B_1(s) = \Big|\gamma \sum_a \pi(a|s) \sum_{s'} [P_{\text{real}}(s'|s,a) - P_{\text{resid}}(s'|s,a)] V(s')\Big|.$$

Pointwise Hölder: for any probability measures P, Q and bounded V, $|E_P V - E_Q V| \le 2\|V\|_\infty \cdot \text{TV}(P, Q)$. Combining with $\|V\|_\infty = \|V^{\pi}_{\mathcal{M}_{\text{real}}}\|_\infty \le R_{\max}/(1-\gamma)$ and restricting the TV supremum to (s, a) pairs in $\text{supp}(d^{\pi}_{\mathcal{M}_{\text{real}}})$ (terms outside the support contribute 0 to the discounted return expectation, see Step 3b of the Lean proof for the formal argument):

$$\sup_s B_1(s) \;\le\; \gamma \cdot \frac{2R_{\max}}{1-\gamma} \cdot \varepsilon.$$

Step 4 (bracket B_2 — Bellman contraction). $B_2(s)$ applies the *same* operator $T^{\pi}_{\mathcal{M}_{\text{resid}}}$ to two different value functions; by Lemma 11, $\sup_s B_2(s) \le \gamma \|V^{\pi}_{\mathcal{M}_{\text{real}}} - V^{\pi}_{\mathcal{M}_{\text{resid}}}\|_\infty = \gamma \|e\|_\infty$.

Step 5 (combine). From Steps 2–4: $\|e\|_\infty \le \frac{2\gamma R_{\max}}{1-\gamma}\varepsilon + \gamma\|e\|_\infty$.

Step 6 (solve). Rearranging, $(1-\gamma)\|e\|_\infty \le \frac{2\gamma R_{\max}}{1-\gamma}\varepsilon$, hence $\|e\|_\infty \le \frac{2\gamma R_{\max}}{(1-\gamma)^2}\varepsilon$. □

G.2 Full proof of Theorem 6

Fix $\gamma = 0.5$. Construct a 2-state 1-action MDP $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a\}$, reward $r(s_1, a) = 0$, $r(s_2, a) = 1$. Set $P_{\text{sim}}(s_1|s_1, a) = P_{\text{sim}}(s_2|s_2, a) = 1$ (self-loops), and $P_{\text{real}}(s_1|s_1, a) = 1$, $P_{\text{real}}(s_1|s_2, a) = 1$, $P_{\text{real}}(s_2|s_2, a) = 0$ (all real trajectories eventually absorb in s_1). Let $\pi_{\text{data}}(a|s_1) = 1$, $\pi_{\text{data}}(a|s_2) = 0$; then $d^{\pi}_{\pi_{\text{data}}}$ concentrates on s_1, so $\text{supp}(d^{\pi}_{\pi_{\text{data}}}) = \{(s_1, a)\}$.

Any $\hat{\delta}$ with $\hat{\delta}(\cdot|s_1, a) = 0$ satisfies $\mathrm{TV}_{\mathrm{supp}(d^{\pi}_{\pi_{\mathrm{data}}})}(P_{\mathrm{real}}, P_{\mathrm{sim}} + \hat{\delta}) = 0$; we choose $\hat{\delta}(s_2|s_2, a) = 1$ so that $\mathcal{M}_{\mathrm{resid}}$ has s_2 as an absorbing state (the same as P_{sim}). Now consider $\pi'(a|s_2) = 1$. Then $V^{\pi'}_{\mathcal{M}_{\mathrm{real}}}(s_2) = 0$ (escapes to s_1 next step, reward 0 forever after) but $V^{\pi'}_{\mathcal{M}_{\mathrm{resid}}}(s_2) = \sum_{t\geq 0} \gamma^t \cdot 1 = 1/(1-\gamma) = 2$. Hence $\|V^{\pi'}_{\mathcal{M}_{\mathrm{real}}} - V^{\pi'}_{\mathcal{M}_{\mathrm{resid}}}\|_\infty \geq 2 > 0$ despite perfect fit on π_{data}'s support. □

G.3 Full proof of Theorem 9

Under Assumption 2, for the Bayes-optimal discriminator φ^* on data pairs $\{(s, a) \mapsto \text{real}, (s, a) \mapsto \text{sim}\}$ we have $\log \frac{\varphi^*(s,a)}{1-\varphi^*(s,a)} = \log \frac{p_{\mathrm{real}}(s'|s,a)}{p_{\mathrm{sim}}(s'|s,a)}$. Substituting two Gaussians of the same variance σ^2, $\log p_{\mathrm{real}} - \log p_{\mathrm{sim}} = (e_{\mathrm{sim}} - e_{\mathrm{real}})/(2\sigma^2)$ where $e_X(s, a) := \|\mu_X(s, a) - s'\|^2$. Rewriting $e_{\mathrm{sim}} = e_{\mathrm{resid}} + \mathrm{const}$ (since $\mathcal{M}_{\mathrm{resid}}$ is a valid kernel close to P_{sim} on support), we obtain the log-odds formula quoted in the main text. The Q^Δ log-target $\log Q^\Delta = -e_{\mathrm{real}}/\tau$ ($\tau = 2\sigma^2$) is direct from the squared-residual definition in Eq. (2) with $\gamma_\Delta = 0$. The difference $\log \frac{\varphi^*}{1-\varphi^*} - \log Q^\Delta = e_{\mathrm{resid}}/\tau$ is a function of e_{resid} alone. Any constant cannot change rankings; a δ-bounded variation can only swap pairs whose $|\log Q^\Delta|$-gap is at most $2\delta/\tau$. □

G.4 Full proof of Theorem 7

The refit operator F factors as the composition $F = F_{\mathrm{reg}} \circ F_{\mathrm{visit}} \circ F_{\mathrm{pol}}$ of three maps: $F_{\mathrm{pol}} : \hat{\delta} \mapsto \pi[\hat{\delta}]$ (policy induction), $F_{\mathrm{visit}} : \pi \mapsto d^\pi$ (visitation), $F_{\mathrm{reg}} : (\hat{\delta}, d^\pi) \mapsto \mathrm{argmin}_{\hat{\delta}'} \mathbb{E}_{(s,a)\sim d^\pi} \|(P_{\mathrm{real}} - P_{\mathrm{sim}} - \hat{\delta}')(\cdot|s, a)\|_2^2$ (L^2 regression). By Assumption 1(a)(b)(c), F_{pol}, F_{visit}, and F_{reg} are Lipschitz with constants L_π, L_d, and B, respectively. Composition of Lipschitz maps gives $\|F(\hat{\delta}_1) - F(\hat{\delta}_2)\|_\infty \leq L_\pi L_d B \, \|\hat{\delta}_1 - \hat{\delta}_2\|_\infty$. Since the residual function space is a closed subset of the finite-dimensional Euclidean space $\mathbb{R}^{|\mathcal{S}|^2|\mathcal{A}|}$ (hence complete under sup-norm), Banach's fixed-point theorem [3] applies whenever $L_\pi L_d B < 1$, giving the unique fixed point $\hat{\delta}^*$ and geometric convergence rate $(L_\pi L_d B)^k$. □

Lean axioms. Only three axioms are used in `ResidualMDP.lean`: (A1) measure-theoretic TV-Hölder identity (`bellman_op_kernel_diff`; provable from standard Mathlib but requires additional setup outside our scaffold); (A2) Bayes-optimal discriminator log-odds identity (`phi_bayes_optimal_logodds`); and (A3) Banach fixed-point theorem for finite-dimensional residual space (`banach_finite_residual`; available in Mathlib as `ContractingWith.fixedPoint` but requires metric-space instance boilerplate). We flag these explicitly as `axiom` in the Lean source. Outside of (A1)–(A3), every `theorem` in `ResidualMDP.lean` closes without `sorry` or `admit`; we do not claim our own contribution to (A1)–(A3).

We refer the reader to the Lean file (supplementary material) for the mechanized proofs. We highlight here only the key Lemma 11.

Lemma 11 (Bellman γ-contraction)**.** *For any finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with $\gamma \in [0, 1)$ and any policy π, $\|T^\pi_{\mathcal{M}} V_1 - T^\pi_{\mathcal{M}} V_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$.*

Proof. For each state s, $(T^\pi V_i)(s) = \sum_a \pi(a|s)[r(s, a) + \gamma \sum_{s'} P(s'|s, a)V_i(s')]$. The difference of the two operators at s is γ times an expectation of $(V_1 - V_2)$ under a probability measure, and an expectation is bounded above by the supremum, giving $|(T^\pi V_1)(s) - (T^\pi V_2)(s)| \leq \gamma \|V_1 - V_2\|_\infty$. Taking sup over s completes the proof. □